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SYSTEM AND METHOD FOR PLANNING OPERATIONS OF LARGE-SCALE AUTONOMOUS VEHICLE FLEET

Abstract

An automated storage and retrieval system includes a storage array, a plurality of autonomous guided bots, and a controller connected to each autonomous guided bot to assign a series of tasks to each autonomous guided bot, which series of tasks includes a task to an autonomous guided bot moving the autonomous guided bot from initial location to a different final location via bot routes. The controller is configured with a bot route planner that has a resolver that seeks conflicts between autonomous guided bots on the bot routes describing bot paths, and resolves each bot route to determine the bot route, at least one bot route being determined based on bot priority, and the resolver sequences the bot routes into a sequence of bot route leg batches, where route legs, forming each bot route in entirety, are divided into corresponding batch of the sequence of route leg batches.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a non-provisional of, and claims the benefit of, U.S. provisional patent application No. 63/558,415 filed on Feb. 27, 2024, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND
1. Field
[0002] The exemplary embodiments generally relate to material handling systems, and more particularly, to transport of items within the material handling system.
2. Brief Description of Related Developments
[0003] Generally multi-agent pathfinding is employed in automated systems, such as logistics facilities and warehouses, to determine collision-free paths for groups of autonomous transport vehicles (i.e., agents) that transport items within the automated system. One example of multi-agent pathfinding is the prioritized planning(PP) technique where fixed priorities of planning goals are found and then plans for the autonomous transport vehicles are generated given these priorities. Another example, of multi-agent pathfinding is the priority-based search (PBS) technique, which does not assume a fixed priority. Rather, priority-based search specifies priorities only on demand. An extension of priority-based search, referred to as the priority-based search with precedence constraint (PBS-PC) technique, may be employed to handle cases where each autonomous transport vehicle has a sequence of goals (i.e., route legs). In the priority-based search techniques (PBS and PBS-PC), the priorities are specified automatically and systematically to resolve planning conflicts and optimize routing performance. However, the priority-based search techniques are centralized multi-agent processes having runtimes that are dominated by the number of conflicts between agents to resolve. In practice, the runtime of the process increases cubically as the number of conflicts to resolve increases. In practice, the number of conflicts to resolve can rapidly increase when the number of tasked autonomous transport vehicles increases or autonomous transport vehicle traffic within the automated system becomes severe (i.e., several autonomous transport vehicles are tasked to travel to or within a common/same region of the logistics facility or warehouse).
[0004] Accordingly, the present disclosure addresses a number of those issues.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] The foregoing aspects and other features of the present disclosure are explained in the following description, taken in connection with the accompanying drawings, wherein:
[0006] FIG. **1** is an exemplary block diagram of an automated storage and retrieval system incorporating aspects of the present disclosure;
[0007] FIG. **1**A is an exemplary schematic illustration of a portion of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0008] FIG. **1**B is an exemplary schematic illustration of a portion of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0009] FIG. **1**C is an exemplary schematic illustration of a portion of the automated storage and

retrieval system of FIG. **1** in accordance with the present disclosure;
[0010] FIG. **1**D is an exemplary schematic illustration of a goods processed by the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0011] FIG. **2**A is an exemplary schematic illustration of a portion of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0012] FIG. **2**B is an exemplary schematic illustration of a portion of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0013] FIG. **2**C is an exemplary schematic illustration of a portion of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0014] FIG. **2**D is an exemplary schematic illustration of a portion of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0015] FIG. **3** is an exemplary schematic illustration of an autonomous guided bot of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0016] FIG. **4** is an exemplary schematic illustration of a portion of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0017] FIG. **4**A is an exemplary schematic illustration of portions of trajectories of an autonomous guided bot of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0018] FIG. **5**A illustrates convention autonomous guided bot routes;
[0019] FIG. **5**B illustrates autonomous guided bot routes of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0020] FIG. **6**A is an exemplary schematic illustration of autonomous guided bot route legs of an autonomous guided bot of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0021] FIG. **6**B is an exemplary schematic illustration of autonomous guided bot route legs for autonomous guided bots of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0022] FIG. **7** is an exemplary autonomous guided bot route leg threat graph for route legs of autonomous guided bots of the automated storage and retrieval system of FIG. **1** in accordance with the present disclosure;
[0023] FIG. **8** is an exemplary threat relation chart in accordance with the present disclosure; and
[0024] FIGS. **9**, **10**, and **11** are exemplary flow diagrams of methods in accordance with the present disclosure.

DETAILED DESCRIPTION

[0025] The following detailed description is meant to assist the understanding of one skilled in the art, and is not intended in any way to unduly limit claims connected or related to the present disclosure.
[0026] The following detailed description references various figures, where like reference numbers refer to like components and features across various figures, whether specific figures are referenced, or not.
[0027] The word “each” as used herein refers to a single object (i.e., the object) in the case of a single object or each object in the case of multiple objects. The words “a,” “an,” and “the” as used herein are inclusive of “at least one” and “one or more” so as not to limit the object being referred to as being in its “singular” form.
[0028] FIG. **1** illustrates an exemplary automated storage and retrieval system **100**, such as of a logistics facility or warehouse, in accordance with aspects of the present disclosure. Although the aspects of the present disclosure will be described with reference to the drawings, it should be understood that the aspects of the present disclosure can be embodied in many forms. In addition, any suitable size, shape or type of elements or materials could be used.
[0029] The present disclosure provides for the automated storage and retrieval system **100**

including a storage array SA with storage locations **130**S arrayed along aisles **130**A and a static (i.e., non-moving) non-deterministic transfer deck **130**DC communicating with each aisles **130**A. The storage array SA includes one or more of the storage array features described herein.

[0030] As also described herein, the storage locations may form breakpack goods interface locations **263**L disposed along aisles formed on or in communication with a static (i.e., non-moving) non-deterministic (goods) transfer deck **130**DG. The storage locations formed by the breakpack goods interface locations **263**L may form a part of the storage array SA or be considered another storage array BSA (with respect to path planning of autonomous guided goods bots **262**) of the automated storage and retrieval system **100** to which the present disclosure is applied independent of path planning of autonomous guided container bots **110**. Path planning of the autonomous guided goods bots **262** and autonomous guided container bots **110** may be performed in conjunction with each other.

[0031] The automated storage and retrieval system **100** includes a plurality of autonomous guided bots or vehicles (e.g., one or more of a plurality of autonomous guided container bots **110** and a plurality of autonomous guided goods bots **262**), each configured for free ranging motion so as to traverse freely along bot paths (e.g., BPT-BPT**3**, see FIGS. **4** and **5**B, as described herein). The bot paths including one or more of time optimal paths, unparameterized paths, and time optimal unparameterized paths, on the non-deterministic deck (e.g., a respective one or more of the container transport deck **130**DC and the breakpack goods deck **130**DG).

[0032] The automated storage and retrieval system **100** includes a controller (such as one or more of control server **120** and warehouse management system **2500** or other suitable controller) that is communicably connected to each autonomous guided bot of the plurality of autonomous guided bots (e.g., one or more of the plurality of autonomous guided container bots **110** and autonomous guided goods bots **262**) so as to assign a series of tasks to each autonomous guided bot. The series of tasks includes at least one task to at least one autonomous guided bot **110**, **262** moving the autonomous guided bot **110**, **262** from an initial location to a different final location via bot routes **499**A-**499**C (see FIGS. **4**, **6**A and **6**B).

[0033] Each of the bot routes **499**A-**499**C may be determined batch by batch in the manner described herein.

[0034] The series of the series of tasks include a current task, and at least one of a preceding task (preceding the current task in series) and a following task (following the current task in series). The at least one task is the current, the preceding task, or the following task.

[0035] The controller is configured with a bot route planner **120**P, **2500**P that has a resolver **120**PR, **2500**PR that seeks (or otherwise determines) conflicts between autonomous guided bots, effecting the series of tasks, on bot routes **499**A-**499**C describing both paths (see FIGS. **4**, **6**A, and **6**B). The resolver **120**PR, **2500**PR resolves each bot route so as to determine the bot route, where at least one bot route is determined based on bot priority. The resolver **120**PR, **2500**PR is arranged to sequence the bot routes into a sequence of bot route legs batches BRL, wherein route legs (as described herein, and illustrated in the Figures as Manhattan distances, which is a metric in which the distance between two points is the sum of the absolute differences in their Cartesian coordinates), forming each bot route in entirety, are divided into corresponding batch of the sequence of route leg batches BRL. It is noted that an autonomous guided bot belonging to a route leg need not be in motion such that a route leg may denote one or more of a traverse path, an initial pose of the autonomous guided bot, and a destination pose of the autonomous guided bot.

[0036] As described herein: the bot route leg batches BRL are prioritized in sequence BRLS, and an earlier or preceding batch in the sequence has a higher priority to a later or subsequent batch in the sequence; at least one bot route leg, in the sequence of bot route legs describing a bot route, is deferred to a later bot route leg batch in the sequence of bot route leg batches; the later bot route leg batch, that includes the deferred at least one bot route leg, is disposed in a later place in the sequence BRLS of bot route leg batches BRL than the at least one bot route leg in sequence of the bot route legs describing the bot route **499**A-**499**C; and/or the deferred at least one bot route leg has a lower priority than undeferred bot route legs.

[0037] The present disclosure provides for, with the controller (e.g., such as one or more of controller **120** and warehouse management system **2500** including the respective resolver **120**PR, **2500**PR, which in some aspects forms part of a control system **100**CS), autonomous transport vehicle travel path planning in the automated storage and retrieval system **100** having one or more large fleet **110**LF, **262**LF of autonomous guided bots (such as autonomous guided container transport vehicles or container bots **110** and/or autonomous goods transport vehicles or autonomous guided goods bots **262** which are collectively and generally referred to herein as autonomous guided bots **110**, **262**).

[0038] Each large fleet **110**LF, **262**LF of autonomous guided bots **110**, **262** includes about three hundred autonomous guided bots **110**, **262**, up to about three hundred autonomous guided bots **110**, **262**, between about twenty autonomous guided bots **110**, **262** and about three hundred autonomous guided bots **110**, **262**, between about twenty autonomous guided bots **110**, **262** and about three hundred autonomous guided bots **110**, **262**, between about forty autonomous guided bots **110**, **262** and about three hundred autonomous guided bots **110**, **262**, between about one hundred autonomous guided bots **110**, **262** and about three hundred autonomous guided bots **110**, **262**, and between about two hundred autonomous guided bots **110**, **262** and about three hundred autonomous guided bots **110**, **262**. While the present disclosure is described herein with respect to non-holonomic autonomous guided bots **110**, **262** (as described herein), the present disclosure is equally applicable with respect to holonomic autonomous transport vehicles or bots such as those produced by Boston Dynamics Inc. of Waltham, Massachusetts (United States) (see, e.g., U.S. Pat. No. 10,265,871 issued on Apr. 23, 2019 and U.S. patent application Ser. No. 17/699,534 filed on Mar. 21, 2022); and those produced by Amazon Technologies Inc. (see, e.g., U.S. Pat. No. 11,643,279 issued on May 9, 2023).

[0039] The automated storage and retrieval system **100** has a control system **100**CS that includes one or more of the control server **120** (also referred to as a controller) and the warehouse management system **2500**. The control system **100**CS is configured (i.e., the one or more of the control server **120** and warehouse management system **2500** has the respective resolver **120**PR, **2500**PR configured) with non-transitory computer program code that embodies a priority-based divide and search autonomous transport vehicle path planning process PBDS (also referred to as a single bot planner) as described herein. The priority-based divide and search autonomous transport vehicle path planning process PBDS is resident in a memory of the control system **100**CS, such as one or more of memory **120**M of control server **120** and memory **2500**M of the warehouse management system **2500**. When the priority-based divide and search autonomous transport vehicle path planning process PBDS is executed by a processor (such as one or more of bot route planner **120**P of control server **120** and bot route planner **2500**P of the warehouse management system **2500**) of the control system **100**CS, the priority-based divide and search autonomous transport vehicle path planning process PBDS causes one or more autonomous transport vehicles **110**, **262** of the automated storage and retrieval system **100** to operate as described herein.

[0040] The priority-based divide and search autonomous transport vehicle path planning process PBDS configures the controller, such as one or more of control server **120** and warehouse management system **2500**, of the automated storage and retrieval system **1000** so that the control controller plans travel paths (which as described herein may be straight paths, arcuate paths, compound paths forming shapes such as “S” shape curves, or any other combination of straight and arcuate paths) of the autonomous transport vehicles **110**, **262** of the respective large fleet **110**LF, **262**LF within several hundred milliseconds, and particularly within about three hundred milliseconds and less than about five hundred milliseconds, and more particularly less than about 100 milliseconds. The autonomous transport vehicles **110**, **262** of each large fleet **110**LF, **262**LF may be isolated from each other so that each large fleet **110**LF, **262**LF is separate and distinct from each other large fleet **110**LF, **262**LF. The large fleets **110**LF, **262**LF operate in separate and distinct areas of the automated storage and retrieval system **100** such that the paths and trajectories of route legs for autonomous container transport vehicles **110** of large fleet **110**LF do not interfere with the paths and trajectories of route legs of the autonomous goods transport vehicles **262** of large fleet **262**LF. Path planning as described herein may be performed for each one of large fleet **110**LF, **262**LF independent of path planning for each other large fleet **110**LF, **262**LF.

[0041] The controller, configured with the priority-based divide and search autonomous transport vehicle path planning process PBDS, divides all route legs RL (e.g., route legs are one or more portions of respective travel paths of respective autonomous transport vehicles **110**, **262** or the respective large fleet **110**LF, **262**LF) of a respective large fleet **110**LF, **262**LF that are to be planned into a sequence BRLS of route leg batches BRL**1**-BRLn (generally referred to herein as "batches BRL"), and plans (e.g., the trajectories and paths of) the route legs RL batch by batch. Dividing all of the route legs RL into batches BRL provides for analysis of a smaller number of route legs RL compared to analysis of all of the route legs RL together, which provides for a fewer number of conflicts (such as for large fleet **110**LF collisions between autonomous transport vehicles **110** and for large fleet **262**LF collisions between autonomous transport vehicles **262**) to resolve in each batch BRL, such that the route legs RL are planned in fewer iterations than if all of the route legs RL were planned together. The sequence BRLS of batch legs BRL prioritizes the batch legs BRL relative to each other from a highest priority to a lowest priority as described herein.

[0042] As will be described in greater detail herein, and with reference to FIG. **1**, for each large fleet **110**LF, **262**LF, the priority-based divide and search autonomous transport vehicle path planning process PBDS of the controller is configured to obtain or otherwise collect unplanned route legs URL from any suitable source such as from one or more vehicle controller VC, where the one or more vehicle controller VC issue(s) at least movement commands to the autonomous transport vehicles **110**, **262** of the automated storage and retrieval system **100**. The one or more vehicle controller VC may be a part of one or more of the controller **120**, warehouse management system **2500**, or otherwise disposed so as to be accessible to the one or more the controller **120** and warehouse management system **2500**. The vehicle controller VC may be a controller **110**C, **262**C of an autonomous transport vehicle **110**, **262**, where the controller **110**C, **262**C generates the route legs in a manner substantially similar to that described in U.S. Pat. No. 11,760,570 issued on Sep. 19, 2023, the disclosure of which is incorporated herein by reference in its entirety. The unplanned route legs URL may be stored in memory **120**M, **2500**M or any other suitable location accessible by the control system **100**CS (the unplanned route legs may be read by the control system **100**CS directly from the one or more vehicle controller VC without storage in memory **120**M, **2500**M).

[0043] For each large fleet **110**LF, **262**LF, the controller, such as one or more of control server **120** and warehouse management system **2500** system, divides all of the unplanned route legs URL into the sequence of batches BRL and plans the trajectories and paths a batch of route legs. Each of the batches BRL are created by the controller by assigning the unplanned route legs URL respective deferral penalties and collecting the unplanned route legs URL into a batch BRL, where each batch BRL has a predetermined maximum number of route legs therein. The predetermined maximum number of route legs may be user-defined maximum number or defined in any other suitable manner. Route legs with higher deferral penalties are collected into and analyzed in batches prior to analyzation of batches including route legs having lower deferral penalties. Dividing the unplanned route legs URL into batches BRL specifies priorities between batches where the priority-based divide and search autonomous transport vehicle path planning process PBDS treats planned trajectories and paths for previously planned route legs as dynamic obstacles for route legs that are planned in a subsequently (lower priority) planned batch BRL of route legs (i.e., all of the route legs planned in previous batches implicitly gain higher priorities with respect to unplanned route legs of unplanned batches).

[0044] The deferral penalty employed, when assigning route legs into a respective batch BRL of the sequence BRLS, represents how much deferring a route leg into a later or subsequent (lower priority) batch BRL impairs the quality of the planning solution. The deferral penalty evaluation of the unplanned route legs URL is based on a notion of what may be referred to as threats (i.e., where threat refers to an effect a higher priority route leg has on a lower priority route leg). Threat graph(s) (see FIG. **7**) may be constructed (FIG. **9**, Block **900**) for employment in the planning process as described herein. For example, where one unplanned route leg URL is put into a subsequent batch relative to another unplanned route leg (i.e., the one unplanned route leg has a lower priority than the other route leg), the one route leg may be delayed, fail to reach its goal, or result in the respective autonomous transport vehicle **110** being unable to remain at a current location, the higher priority route leg (in this example the other route leg) is considered a threat to the lower priority route leg (in this example, the one route leg). The threats may be qualitatively divided into three types (e.g., delay, reachability, and safety) where the three types of threats define different levels of threat severity. With the notion of threats, a threat graph (see FIG. **7**—as described herein) may be constructed and the route leg deferral penalty may be evaluated by the control system **110**CS as a severity of being threatened by route legs in a batch of route legs that is currently being planned (i.e., a current batch of route legs).

[0045] The controller, such as one or more of control server **120** and warehouse management system **2500**, determines if all route legs have been planned (FIG. **9**, Block **910**). Where all route legs have been planned, the planning solution is returned (FIG. **9**, Block **920**) and the corresponding autonomous transport vehicle(s) **110**, **262** are commanded by the controller **120**, **2500** to execute the plan(s). Where all route legs are not planned, the controller **120**, **2500** plans later or subsequent batches BRL of route legs (FIG. **9**, Blocks **930** and **940**), for the respective large fleet **110**LF, **262**LF, avoiding the trajectories and paths of the route legs (whose trajectories and paths were) planned in previously planned batches BRL of route legs for the respective large fleet **110**LF, **262**LF (i.e., as noted above, path planning for each large fleet is performed independent of path planning for each other large fleet). Planning may continue in a loop at least until all route legs have been planned (see FIG. **9**).

[0046] Referring again to FIG. **1**, in accordance with the present disclosure the automated storage and retrieval system **100** may operate in a retail distribution center, warehouse, or the back of a retail store. The automated storage and retrieval system **100** may operate to, for example, fulfill orders received from retail stores for case units such as those described in U.S. Pat. No. 10,822,168 issued on Nov. 3, 2020 and U.S. patent application Ser. No. 17/358,383 filed on Jun. 25, 2021, the disclosures of which are incorporated by reference herein in their entireties. For example, the case units are cases or units of goods not stored in trays, on totes or on pallets (e.g. uncontained). In other examples, the case units are cases or units of goods that are contained in any suitable manner such as in trays, on totes, in containers (such as containers of remainder goods after breakpack where the broken down case unit structure is unsuitable for transport of the remainder goods as a unit) or on pallets. In still other examples, the case units are a combination of uncontained and contained items. It is noted that the case units, for example, include cased units of goods (e.g. case of soup cans, boxes of cereal, etc.) or individual goods that are adapted to be taken off of or placed on a pallet. In accordance with the present disclosure, shipping cases for case units (e.g. cartons, barrels, boxes, crates, jugs, or any other suitable device for holding case units) may have variable sizes and may be used to hold case units in shipping and may be configured so they are capable of being palletized for shipping or sent to a downstream logistics process (e.g., such as goods to person automation) without being palletized. The case units may be segmented case units that include multiple order profiles in one case unit (e.g., such as a segmented tote). The segmented case unit may increase the product density within the case unit and any downstream logistics (e.g., downstream packaging solution such as the goods to person automation). It is noted that when, for example, bundles or pallets of case units arrive at the storage and retrieval system the content of each pallet may be uniform (e.g. each pallet holds a predetermined number of the same item—one pallet holds soup and another pallet holds cereal) and as pallets leave the storage and retrieval system the pallets may contain any suitable number and combination of different case units (e.g. a mixed pallet where each mixed pallet holds different types of case units—a pallet holds a combination of soup and cereal) that are provided to, for example the palletizer in a sorted arrangement for forming the mixed pallet. In the present disclosure the storage and retrieval system **100** described herein may be applied to any environment in which case units are stored and retrieved.

[0047] Also referring to FIG. **1**D, it is noted that when, for example, incoming bundles or pallets (e.g. from manufacturers or suppliers of case units arrive at the storage and retrieval system for replenishment of the automated storage and retrieval system **100**, the content of each pallet may be uniform (e.g. each pallet holds a predetermined number of the same item-one pallet holds soup and another pallet holds cereal). The cases of such pallet load may be substantially similar or in other words, homogenous cases (e.g. similar dimensions), and may have the same SKU (otherwise, as noted before the pallets may be "rainbow" pallets having layers formed of homogeneous cases). As pallets PAL leave the storage and retrieval system **100**, with cases filling replenishment orders, the pallets PAL may contain any suitable number and combination of different case units CU (e.g., each pallet may hold different types of case units-a pallet holds a combination of canned soup, cereal, beverage packs, cosmetics and household cleaners). The cases combined onto a single pallet may have different dimensions and/or different SKU's. In one aspect of the present disclosure, the storage and retrieval system **100** may be configured to generally include an in-feed section, a storage and sortation section (where, in one aspect, storage of items is optional) and an output section as will be described in greater detail below. In the present disclosure the system **100** operating for example as a retail distribution center may serve to receive uniform pallet loads of cases, breakdown the pallet goods or disassociate the cases from the uniform pallet loads into independent case units handled individually by the system, retrieve and sort the different cases sought by each order into corresponding groups, and transport and assemble the corresponding groups of cases into what may be referred to as mixed case pallet loads MPL. In the present disclosure the system **100** operating for example as a retail distribution center may serve to receive uniform pallet loads of cases, breakdown the pallet goods or disassociate the cases from the uniform pallet loads into independent case units handled individually by the system, retrieve and sort the different cases sought by each order into corresponding groups, and transport and sequence the corresponding groups of cases in the manner described in U.S. Pat. No. 9,856,083 issued on Jan. 2, 2018, the disclosure of which is incorporated herein by reference in its entirety.

[0048] The storage and sortation section includes a multilevel automated storage system that has an automated transport system that in turn receives or feeds individual cases into the multilevel storage array SA for storage in a storage area (such as storage spaces **130**S of the storage structure **130**). The storage and sortation section may define outbound transport of case units from the multilevel storage array such that desired case units are individually retrieved in accordance with commands generated in accordance to orders entered into a warehouse management system, such as warehouse management system **2500**, for transport to the output section. The storage and sortation section may receive individual cases, sort the individual cases (utilizing, for example, the buffer and interface stations described herein), e.g., in a case level sortation, and transfer the individual cases to the output section in accordance to orders entered into the warehouse management system. The sorting and grouping of cases according to order (e.g. an order out sequence) may be performed in whole or in part by either the storage and retrieval section or the output section, or both, the boundary between being one of convenience for the description and the sorting and grouping being capable of being performed any number of ways. The intended result is that the output section assembles the appropriate group of ordered cases, that may be different in SKU, dimensions, etc. into mixed case pallet loads in the manner described in, for example, U.S. Pat. No. 8,965,559 issued on Feb. 24, 2015, the disclosure of which is incorporated herein by

reference in its entirety.
[0049] The output section generates the pallet load in what may be referred to as a structured architecture of mixed case stacks. The structured architecture of the pallet load described herein is representative and in other aspects, the pallet load may have any other suitable configuration. For example, the structured architecture may be any suitable predetermined configuration such as a truck bay load or other suitable container or load container envelope holding a structural load. The structured architecture of the pallet load may be characterized as having several flat case layers L**121**-L**125**, L**12**T as described in U.S. Pat. No. 9,856,083, previously incorporated by reference herein in its entirety.
[0050] Referring again to FIG. **1**, the automated storage and retrieval system **100** may include a storage array (e.g., storage structure **130** having storage spaces **130**S) with at least one elevated storage level **130**L and at least one breakpack module **266** (as described herein). It is noted that while the storage array is described as a three dimensional storage array, the storage array may be a two dimensional storage array (e.g., single level floor), the back of a truck, or any other suitable storage array where case units may be transferred directly by the storage and retrieval system **100** (such as by the autonomous guided container bots **110**) or indirectly (e.g., by fork trucks or other vehicle/operator placing case units on a conveyor in a predetermined sequence (grouped stock keeping units or other categorical sequencing)) to a breakpack module **266**. Where the storage array is a single level (i.e., single level floor) the breakpack module **266** is located on the floor level of the storage array. Mixed product units are input and distributed in the storage array in cases CU of product units of common kind per case CU (each case input to the system **100** holds a common kind of stock keeping unit (SKU)). For example, the automated storage and retrieval system **100** includes input stations **160**IN (which include depalletizers **160**PA and/or conveyors **160**CA for transporting items (e.g., inbound supply containers) to lift modules **150**A for entry into a storage level **130**L of the storage structure **130**).
[0051] The automated storage and retrieval system **100** includes an automated transport system (e.g., autonomous container transport vehicles or autonomous guided container bots **110**, autonomous goods transport vehicles or autonomous guided goods bots **262**, breakpack modules, and other suitable level transports described herein) with at least one asynchronous transport system for transporting cases/products on a given storage structure level **130**L (e.g., level transport). As will be described, the storage and retrieval system **100** may include non-holonomic autonomous guided container bots **110** that undeterministically (i.e., are not physically constrained for travel along a given path, not restricted to Cartesian motion, and not restricted to travel lanes of a Cartesian grid of travel lanes) travel along one or more physical pathways (such as described with respect to FIGS. **4**-**5**B) of the storage and retrieval system **100** to provide at least one level of asynchronicity. The autonomous guided container bots **110** of each respective storage level **130**L may be confined to the respective storage level **130**L such that each storage level has a respective large fleet **110**LF of autonomous guided container bots **110** for which path planning is determined as described herein independent of other large fleets **110**LF of other storage levels **130**L. The storage structure may be configured such that autonomous guided container bots **110** may travel between two or more storage levels **130**L such that the large fleet **110**LF includes the autonomous guided container bots **110** of the two or more storage levels **130**L. The autonomous guided container bots **110** are configured for high speed travel, where the high speed travel is greater than about 1 m/sec or more with the container bot **110** carrying a payload of about 60 lbs (about 27 kg) to about 90 lbs (about 41 kg) (in other aspects the payload may be less than about 60 lbs or more than about 90 lbs). In accordance with another example, the high speed travel of the container bot **110** may be in excess of about 20 km/hr (e.g. about 5.6 m/sec) and more particularly about 32 km/hr (e.g. about 9.144 m/sec) or about 36 km/hr (e.g. about 10 m/sec) with the container bot **110** carrying a payload of about 60 lbs (about 27 kg) to about 90 lbs (about 41 kg) (in other aspects the payload may be less than about 60 lbs or more than about 90 lbs).

[0052] At least another level of asynchronicity is provided (as described herein) such that, for example, case/product holding locations are greater than the number of bots transporting cases/products. At least one lift **150** is provided for transporting cases/products between storage levels (e.g., between level transport) or the cases/products may be presorted an on a predetermined level before a container bot **110** retrieves the cases/products (e.g., such that the lift does not transfer the cases/products between levels for container bot **110** retrieval). The at least one lift **150**B is communicably connected to the storage array as described herein so as to automatically retrieve and output, from the storage array, product units distributed in the cases in a common part (e.g., the storage locations **130**S of a respective storage level **130**L) of the at least one elevated storage level **130**L of the storage array. The output product units being one or more of mixed singulated product units, in mixed packed groups, and in mixed cases. The automated storage and retrieval system **100** may include output stations **160**UT, **160**EC (which include palletizers **160**PB, operator stations **160**EP and/or conveyors **160**CB for transporting items (e.g., outbound supply containers and filled breakpack goods (order) containers) from lift modules **150**B for removal from storage (e.g., to a palletizer (for palletizer load) or to a truck (for truck load)). Here the output station **160**EC is an individual fulfillment (or e-commerce) output station where, for example, filled breakpack goods (order) containers including single goods items and/or small bunches of goods are transported for fulfilling an individual fulfillment order (such as an order placed over the Internet by a consumer). The output station **160**UT is a commercial output station where large numbers of goods are generally provided on pallets for fulfilling orders from commercial entities (e.g., commercial stores, warehouse clubs, restaurants, etc.). The automated storage and retrieval system **100** may include both the commercial output station **160**UT and the individual fulfillment output station **160**EC; although, the automated storage and retrieval system may include one or more of the commercial output station **160**UT and the individual fulfillment output station **160**EC.

[0053] The automated storage and retrieval system **100** also includes the input and output vertical lift modules **150**A, **150**B (generally referred to as lift modules **150**—it is noted that while input and output lift modules are shown, a single lift module may be used to both input and remove case units from the storage structure), a storage structure **130** (which may have at least one elevated storage level as noted above and in some aspects forms a multilevel storage array), and at least one autonomous guided container bot **110** (which form at least a part of the asynchronous transport system for level transport) which may be confined to a respective storage level of the storage structure **130** and are distinct from a container transfer deck **130**DC on which they travel. The lift modules **150** include any suitable transport configured to vertically raise and lower case units and are inclusive of reciprocating elevator type lifts, fork lift trucks, etc. It is noted that the depalletizers **160**PA may be configured to remove case units from pallets so that the input station **160**IN can transport the items to the lift modules **150** for input into the storage structure **130**. The palletizers **160**PB may be configured to place items removed from the storage structure **130** on pallets PAL (FIG. **1**D) for shipping. As used herein the lift modules **150**, storage structure **130** and autonomous guided container bots **110** may be collectively referred to herein as the multilevel automated storage system (e.g. storage and sorting section) noted above so as to define (e.g. relative to e.g. a container bot **110** frame of reference REF—FIG. **4**A—or any other suitable storage and retrieval system frame of reference) transport/throughput axes (in e.g. three dimensions) that serve the three dimensional multilevel automated storage system where each throughput axis has an integral “on the fly sortation” (e.g. sortation of case units during transport of the case units) so that case unit sorting and throughput occurs substantially simultaneously without dedicated sorters as described in U.S. Pat. No. 9,856,083, previously incorporated herein by reference in its entirety.

[0054] Also referring to FIGS. **1**, **1**C, **2**A, and **2**C, the storage structure **130** may include a container autonomous transport travel loop(s) **233**, **233**A (e.g., formed on and along a container transfer deck **130**DC), disposed at a respective level of the storage structure **130**. It is noted that the lifts **150** are connected via transfer stations TS (also referred to herein as container infeed stations when the lift

150 is an inbound lift **150**A or as container outfeed stations when the lift **150** is an outbound lift **150**B) to the container transfer deck **130**DC, and each lift is configured to lift one or both of supply containers **265** (empty or filled) (see FIG. **2**C) and the breakpack goods containers **264** (empty or filled) (see FIG. **2**C) into and out of the at least one elevated storage level **130**L of the storage structure **130**. Container storage locations (or spaces) **130**S are arrayed peripherally along the container transfer deck **130**DC. For example, multiple storage rack modules RM, configured in a high-density three dimensional rack array RMA, are accessible by storage or deck levels **130**L. As used herein the term "high density three dimensional rack array" refers to the three dimensional rack array RMA having undeterministic open shelving distributed along picking aisles **130**A where, multiple stacked shelves may be accessible from a common picking aisle travel surface or picking aisle level as described in U.S. Pat. No. 9,856,083, previously incorporated by reference herein in its entirety.

[0055] Each storage level **130**L includes pickface storage/handoff spaces **130**S (referred to herein as storage spaces **130**S or container storage locations **130**S) arrayed peripherally along the container transfer deck **130**DC. At least one of the storage locations **130**S is a supply container storage location **130**SS, and another of the container storage locations is a breakpack goods (or order) container storage location **130**SB. The storage spaces **130**S are in one aspect formed by the rack modules RM where the rack modules include shelves that are disposed along storage or picking aisles **130**A (that are connected to the container transfer deck **130**DC) which, e.g., extend linearly through the rack module array RMA and provide container bot **110** access to the storage spaces **130**S and transfer deck(s) **130**B. The shelves of the rack modules RM may be arranged as multi-level shelves that are distributed along the picking aisles **130**A. The autonomous guided container bots **110** travel on a respective storage level **130**L along the picking aisles **130**A and the container transfer deck **130**DC for transferring case units between any of the storage spaces **130**S of the storage structure **130** (e.g. on the level which the container bot **110** is located) and any of the lift modules **150** (e.g. each of the autonomous guided container bots **110** has access to each storage space **130**S on a respective level and each lift module **150** on a respective storage level **130**L). The transfer decks **130**B are arranged at different levels (corresponding to each level **130**L of the storage and retrieval system) that may be stacked one over the other or horizontally offset, such as having one container transfer deck **130**DC at one end or side RMAE**1** of the storage rack array RMA or at several ends or sides RMAE**1**, RMAE**2** of the storage rack array RMA as described in, for example, U.S. patent application Ser. No. 13/326,674 filed on Dec. 15, 2011 the disclosure of which is incorporated herein by reference in its entirety.

[0056] The container transfer decks **130**DC are substantially open and configured for the undeterministic traversal of autonomous guided container bots **110** along multiple travel lanes (e.g. along an X throughput axis with respect to the bot frame of reference REF illustrated in FIG. **4**A) across and along the transfer decks **130**B. As will be described in further detail below (and as described in U.S. Pat. No. 10,556,743 issued on Feb. 11, 2020 and having application Ser. No. 15/671,591, the disclosure of which is incorporated herein by reference in its entirety) the multiple travel lanes may be configured to provide multiple access paths or routes to each storage location **130**S (e.g., pickface, case unit, container, or other items stored on the storage shelves of rack modules RM) so that autonomous guided container bots **110** may reach each storage location using, for example, a secondary path if a primary path to the storage location is obstructed. The transfer deck(s) **130**B at each storage level **130**L communicate with each of the picking aisles **130**A on the respective storage level **130**L. Autonomous guided container bots **110** bi-directionally traverse between the container transfer deck(s) **130**DC and picking aisles **130**A on each respective storage level **130**L so as to travel along the picking aisles (e.g. along the X throughput axis with respect to the bot frame of reference REF illustrated in FIG. **4**A) and access the storage spaces **130**S disposed in the rack shelves alongside each of the picking aisles **130**A (e.g. autonomous guided container bots **110** may access, along a Y throughput axis, storage spaces **130**S distributed on both sides of each aisle such that the container bot **110** may have a different facing when traversing each picking aisle **130**A, for example, referring to FIG. **4**A, drive wheels **202** leading a direction of travel or drive wheels trailing a direction of travel). Throughput outbound from the storage array in the horizontal plane corresponding to a predetermined storage or deck level **130**L is effected by and manifest in the combined or integrated throughput along both the X and Y throughput axes. As noted above, the container transfer deck(s) **130**DC may provide container bot **110** access to each of the lifts **150** on the respective storage level **130**L where the lifts **150** feed and remove case units (e.g. along the Z throughput axis) to and/or from each storage level **130**L and where the autonomous guided container bots **110** effect case unit transfer between the lifts **150** and the storage spaces **130**S.

[0057] As described above, referring also to FIG. **2**A, the storage structure **130** may include multiple storage rack modules RM, configured in a three dimensional array RMA where the racks are arranged in aisles **130**A, the aisles **130**A being configured for container bot **110** travel within the aisles **130**A. The container transfer deck **130**DC has an undeterministic transport surface on which the autonomous guided container bots **110** travel where the undeterministic transport surface (also referred to herein as a deck surface) **130**BS has multiple travel lanes (e.g., more than one juxtaposed travel lane (e.g. high speed bot travel paths HSTP)) for travel of the container bot **110** along the container autonomous transport travel loop(s) **233**, **233**A formed by the container transfer deck **130**DC, where the multiple travel lanes connect the aisles **130**A. The container autonomous transport travel loop **233**A provides the container bot **110** with random access to any and each picking aisle **130**A and random access to any and each lift **150**A, **150**B on the respective level **130**L of the storage structure **130**. At least one of the multiple travel lanes has a travel sense opposite to another travel lane sense of another of the multiple travel lanes (so as to form the container autonomous transport travel loop **233**).

[0058] Any suitable controller of the storage and retrieval system **100** such as for example, control server **120**, may be configured to create any suitable number of alternative pathways for retrieving one or more case units (and/or breakpack containers) from their respective storage locations **130**S when a pathway provided access to those case units is restricted or otherwise blocked. For example, the control server **120** may include suitable programming, memory and other structure for analyzing the information sent by the container **110**, lifts **150**A, **150**B, and input/output stations **160**IN, **160**UT, **160**EC for planning a container bot's **110** primary or preferred route to a predetermined item within the storage structure. The preferred route may be the fastest and/or most direct route that the container bot **110** can take to retrieve the case units/pickfaces. The preferred route may be any suitable route. The control server **120** may also be configured to analyze the information sent by the autonomous guided container bots **110**, the lifts **150**A, **150**B, and input/output stations **160**IN, **160**UT, **160**EC for determining if there are any obstructions along the preferred route. If there are obstructions along the preferred route the control server **120** may determine one or more secondary or alternate routes for retrieving the case units so that the obstruction is avoided and the case units can be retrieved without any substantial delay in, for example, fulfilling an order. It should be realized that the container bot route planning may also occur on the container bot **110** itself by, for example, any suitable control system, such as a controller (system) **110**C onboard the container bot **110**. As an example, the bot control system may be configured to communicate with the control server **120** for accessing the information from other autonomous guided container bots **110**, the lifts **150**A, **150**B, and the input/output stations **160**IN, **160**UT, **160**EC for determining the preferred and/or alternate routes for accessing an item in a manner substantially similar to that described above. It is noted that the container bot **110** controller **110**C may include any suitable programming, memory and/or other structure to effect the determination of the preferred and/or alternate routes.

[0059] Referring to FIG. **2**A, as a non-limiting example, in an order fulfillment process the container bot **110**A, which is traversing container transfer deck **130**DC, may be instructed to retrieve item **499** from picking aisle **131**. However, there may be a disabled bot **110**B blocking aisle **131** such that the bot **110**A cannot take a preferred (e.g. the most direct and/or fastest) path to the case unit **499**. In this example, the control server **120** may instruct the container bot **110**A to traverse an alternate route such as through any unreserved picking aisle (e.g. an aisle without a container bot in it or an aisle that is otherwise unobstructed) so that the container bot **110**A can travel along, for example, another container transfer deck **130**DC**2** that is substantially similar to container transfer deck **130**DC. The container bot **110**A can enter the end of the picking **131**, opposite the blockage, from the other container transfer deck **130**DC**2** so as to avoid the disabled container bot **110**B for accessing the item **499**. In another aspect, the storage and retrieval system **100** may include one or more bypass aisles **132** that run substantially transverse to the picking aisles **130** to allow the autonomous guided container bots **110** to move between picking aisles **130** in lieu of traversing the container transfer decks **130**DC, **130**DC**2**. The bypass aisles **132** may be substantially similar to travel lanes of the container transfer decks **130**DC, **130**DC**2**, as described herein, and may allow bidirectional or unidirectional travel of the autonomous guided container bots through the bypass aisle **132**. The bypass aisle **132** may provide one or more lanes of container bot travel where each lane has a floor and suitable guides for guiding the bot along the bypass aisle **132** in a manner similar to that described herein with respect to the transfer decks **130**DC, **130**DC**2**. The bypass aisles **132** may have any suitable configuration for allowing the autonomous guided container bots **110** to traverse between the picking aisles **130**. It is noted that while the bypass aisle **132** is shown with respect to a storage and retrieval system having transfer decks **130**DC, **130**DC**2** disposed on opposite ends of the storage structure, in other aspects, a storage and retrieval system **100** having only one transfer deck may also include one or more bypass aisles **132**.

[0060] A breakpack module **266**AL may be located on a side of the container transfer deck **130**DC on which the picking aisles **130** are located and one or more picking aisles **130** extend into the breakpack module **266**AL so as to form container bot riding surface(s) **266**RS. Here the container bot **110**A is to deliver a supply container **265** to the breakpack module **266**AL and the picking aisle **133** extending into the breakpack module is blocked by container bot **110**D. The control server **120** and/or container bot controller **110**C may determine a secondary or bypass route for the container bot **110**A to access breakpack station (either travelling along the other container transfer deck **130**DC**2** and/or bypass aisle **132**) in a manner substantially similar to that described above with respect to item **499**.

[0061] It is noted that the storage and retrieval systems shown and described herein have exemplary configurations only and in other aspects the storage and retrieval systems may have any suitable configuration and components for storing and retrieving items as described herein. For example, the storage and retrieval system may have any suitable number of storage sections, any suitable number of transfer decks, any suitable number of breakpack modules, and corresponding input/output stations.

[0062] The juxtaposed travel lanes are juxtaposed along a common undeterministic transport surface **130**BS between opposing sides **130**BD**1**, **130**BD**2** of the container transfer deck **130**DC. As illustrated in FIG. **2**A, the aisles **130**A may be joined to the container transfer deck **130**DC on one side **130**BD**2** of the container transfer deck **130**DC although, the aisles may be joined to more than one side **130**BD**1**, **130**BD**2** of the container transfer deck **130**DC in a manner substantially similar to that described in U.S. patent application Ser. No. 13/326,674 filed on Dec. 15, 2011, the disclosure of which is previously incorporated by reference herein in its entirety. As will be described in greater detail below the other side **130**BD**1** of the container transfer deck **130**DC may include includes deck storage racks (e.g. interface stations (also referred to as transfer stations) TS and buffer stations BS) that are distributed along the other side **130**BD**1** of the container transfer deck **130**DC so that at least one part of the transfer deck is interposed between the deck storage racks (such as, for example, buffer stations BS or transfer stations TS) and the aisles **130**A. The deck storage racks are arranged along the other side **130**BD**1** of the container transfer deck **130**DC so that the deck storage racks communicate with the autonomous guided container bots **110** from the container transfer deck **130**DC and with the lift modules **150** (e.g. the deck storage racks are accessed by the autonomous guided container bots **110** from the container transfer deck **130**DC and by the lifts **150** for picking and placing pickfaces so that pickfaces are transferred between the autonomous guided container bots **110** and the deck storage racks and between the deck storage racks and the lifts **150** and hence between the autonomous guided container bots **110** and the lifts **150**).

[0063] Referring again to FIG. **1**, each storage level **130**L may include charging stations **130**C for charging an on-board power supply of the autonomous guided container bots **110** on that storage level **130**L such as described in, for example, U.S. patent application Ser. No. 14/209,086 filed on Mar. 13, 2014 and Ser. No. 13/326,823 filed on Dec. 15, 2011 (now U.S. Pat. No. 9,082,112), the disclosures of which are incorporated herein by reference in their entireties.

[0064] Referring to FIGS. **1**, **2**A, **2**C, as noted above, the automated storage and retrieval system **100** may include one or more break pack modules **266**. The breakpack modules **266** are configured to break down product containers or case units CU into breakpack goods containers **264** for order fulfillment, such as described in U.S. patent application Ser. No. 17/358,383 filed Jun. 25, 2021, the disclosure of which was previously incorporated herein by reference in its entirety. The breakpack modules **266** may operate as an automated decant process for downstream logistics such as goods to person automation. One or more breakpack modules **266** may be located on a common level **130**L of the automated storage and retrieval system, where one or more levels of the automated storage and retrieval system **100** include at least one breakpack module **266**. The breakpack module(s) **266** may be plug and play modules that may be coupled to any suitable portion of the structure of the automated storage and retrieval system **100**. For example, the breakpack module(s) may be coupled to a container transfer deck **130**DC or picking (or pick) aisle(s) **130**A of the automated storage and retrieval system **100** as will be described in greater detail below. The breakpack module(s) **266** may be disposed on any suitable number of stacked storage levels of the automated storage and retrieval system **100**. Here, the automated storage and retrieval system **100** may be configured, such as through any suitable controller (e.g., control server **120**) so that the automated storage and retrieval system **100** has selectable modes of operation. In one mode of operation the automated storage and retrieval system **100** is configured to output product cases, containers, and/or case units to a palletizer. In another mode of operation, such as with the breakpack module(s) **266** employed, the automated storage and retrieval system **100** is configured to break down product cases, product containers, and/or case units and output breakpack goods containers, product cases, containers, and/or case units to a palletizer, or in other aspects, re-enter the breakpack (order) container(s) and/or a remainder of a product cases, containers, and/or case units to a palletizer (e.g., after being broken down) into storage for later retrieval.

[0065] The controller **120** (or warehouse management system **2500**), may be configured to plan travel paths and effect operation of a container bot **110** and an autonomous guided goods bot **262** (both of which form at least part of the asynchronous transport system) (see also, e.g., FIG. **2**C) for assembling orders of breakpack goods BPG from supply containers **265** into breakpack goods containers **264** and outfeed of breakpack goods containers **264** through container outfeed stations TS as will be described herein. The controller **120** (or warehouse management system **2500**) may be configured to effect operation of the container bot(s) **110** between the container storage locations **130**S, the breakpack operation station **140**, and a breakpack goods container **264** located along the breakpack goods transfer deck **130**DG. As another example, the controller **120** is configured to effect operation of the autonomous guided goods bot(s) **262** so that transport of the breakpack goods BPG, by the autonomous guided goods bot **262** traverse on the goods transfer deck **130**DG, sorts the breakpack goods BPG, e.g., in a unit/each level sortation, to corresponding breakpack goods containers **264**. The controller **120** may be configured to effect operation of the container bot(s) **110** so that the container bot(s) **110** accesses corresponding breakpack goods containers **264** at the goods transfer deck **130**DG and transports the breakpack goods containers **264** via traverse along the container transfer deck **130**DC to at least one of a container output/transfer station TS and a corresponding container storage location **130**SB of storage shelves of a corresponding level **130**L of the multilevel storage array.

[0066] The controller **120** (or warehouse management system **2500**) may be configured to plan travel paths and effect operation of the container bot(s) **110** and effect operation of lifts **150** (e.g., to form a container supply system) so as to introduce empty breakpack goods containers **264** into the automated storage and retrieval system so that the container bot(s) **110** transport the empty breakpack goods containers **264**, along the transport loops **233**, **233**A of the container transfer deck(s) **130**DC and into a breakpack module **266** for placement at a breakpack goods interface location(s) **263**L of a breakpack goods interface **263** for transfer of breakpack goods BPG into the breakpack goods containers **264**. Empty breakpack containers **264** may be transferred to (in a manner similar to that noted above with the lifts and autonomous guided container bots) and stored in the storage spaces **130**SB, **130**S of the rack modules RM or buffered at an infeed station, where the controller **120** is configured to effect transfer of the empty breakpack goods containers **264** from the storage spaces **130**SB, **130**S or buffer location to the breakpack goods interface **263** in a manner similar to that described above. The controller **120** may be configured to effect operation of the container bot(s) **110** and lifts **150** (e.g., forming a container supply system) so as to introduce empty supply containers **265** or standardized containers **265**S (as described herein) into the automated storage and retrieval system so that the container bot(s) **110** transport the empty supply containers **265** or standardized containers **265**S, along the transport loops **233**, **233**A of the container transfer deck(s) **130**DC and to the breakpack operation station **140** of a breakpack or directly or indirectly to a downstream logistics process such as the goods to person process.

[0067] Each breakpack module **266** may have a container bot riding surface **266**RS that forms a portion **130**DCP of the container transfer deck **130**DC, where the riding surface **266**RS is substantially similar to that of container transfer deck **130**DC, while in other aspects the container bot riding surface **266**RS may be substantially similar to that of the picking aisles **130**A. For ease of explanation, the aspects of the present disclosure will refer to the container bot riding surface **266**RS within the breakpack module **266** as a portion of the container transfer deck **130**DC. Where the bot riding surface **266**RS is formed by a portion of (or is an extension of) the container transfer deck **130**DC it is noted that, while the container transfer deck **130**D is illustrated in FIG. **2**C a single path transport loop, in other aspects the transport loop of the breakpack module **266** may be a multilane transport loop substantially similar to container transport deck illustrated in FIG. **2**A. For example, referring to FIG. **2**E the container bot travel surface **266**RS is an open undeterministic travel surface having multiple travel inbound and outbound lanes. For example, there are multiple inbound travel lanes TL**1**, TL**2** where travel lane TL**2** is a bypass lane for travelling around obstructions on travel lane TL**1** (or vice versa). There may also be multiple outbound travel lanes TL**3**, TL**4**, TL**5**. Here, travel lane TL**5** defines a queue lane **130**QL (FIG. **2**C) for the autonomous guided container bots **110** at the breakpack goods interface **263** while travel lanes TL**4** and TL**5** may be used for egress from the breakpack module **266**, with travel lane TL**5** being a bypass for travelling around obstructions on travel lane TL**4** (or vice versa). Case units may be transferred between the storage array and the breakpack module **266** indirectly, such as by conveyors and/or fork trucks. In one or more aspects, the autonomous guided container bots **110** or fork trucks may deliver case units to conveyors that transport the case units to the breakpack station and from the breakpack station to the storage array or downstream logistics process.

[0068] Each of the breakpack modules **266** includes a breakpack goods autonomous transport travel loop **234** (see exemplary breakpack goods autonomous transport travel loops **234**A-**234**E formed on and along a goods deck or goods transfer deck **130**DG), at least one breakpack operation station **140**, and a breakpack goods interface **263** disposed between and interfacing the goods transfer deck **130**DG with the container transfer deck **130**DC. Referring also to FIGS. **1** and **2**A, the breakpack goods module **266** may include one or more belt sorters BST (such as cross belt sorters) that is/are configured as an interface(s) between autonomous guided goods bots **262** (operating on the goods deck **130**DG) and the autonomous guided container bots **110** (operating on the container transport deck **130**DC), between the autonomous guided container bots **110** and the breakpack operation station **140**, and/or between the breakpack operation station **140** and the autonomous guided goods bots **262**. For exemplary purposes only, the goods deck **130**DG is illustrated as having three travel lanes that form the (variable length) travel loops **234**A-**234**E; however, the goods deck may have any suitable number of travel lanes that form any suitable number of breakpack goods autonomous transport travel loops **234**. Each breakpack module **266** may be undeterministically coupled (e.g., the breakpack modules **266** maybe coupled to the automated storage and retrieval system **100** at any suitable location thereof, such as to one or more ends **130**BE**1**, **130**BE**2**, or centrally located between the two ends **130**BE**1**, **130**BE**2** such as in place of picking aisles **130** (and storage locations) or at any other suitable location) to the automated storage and retrieval system **100** in any suitable manner (e.g., so as to form a part thereof). Though the breakpack modules **266** are coupled undeterministically to the structure of the automated storage and retrieval system **100** each component of the breakpack modules **166** is independent (e.g., self-contained as a unit) and/or independently automated in guidance and travel of the bots (e.g., autonomous guided goods bots **262**) from the components of the automated storage and retrieval system, so that the interface between the components of the breakpack modules **266** and the components of the automated storage and retrieval system **100** is undeterministic.

[0069] The breakpack module(s) **266** may be coupled to the structure of the automated storage and retrieval system **100** at any suitable location and at any suitable level(s) **130**L. For example, as noted above, a break pack module **266** may be located at one or more ends **130**BE**1**, **130**BE**2** of the container transfer deck **130**DC or at one or more sides **130**BD**1**, **130**BD**2** of the container transfer deck **130**DC (such as in lieu of storage rack modules RM/picking aisles **130**A or lifts **150**A, **150**B, or as an extension of one or more picking aisles **130**A). Each of the breakpack modules **266** is a plug and play module that is integrated with (or otherwise connected to) the container transfer deck **130**DC so that the container transfer deck **130**DC is communicably coupled to the container bot riding surface **266**RS. The container transfer deck **130**DC may extend into the breakpack module to form the container bot riding surface **266**RS (e.g., the breakpack module forms a modular part of the container transfer deck **130**DC) so that autonomous guided container bots **110** traverse or move into and out of the breakpack modules **266** along the undeterministic container transfer deck **130**DC, and at least one of the multiple travel lanes of the container transfer deck **130**DC defines a queue lane **130**QL (FIG. **2**C) for the autonomous guided container bots **110** at the breakpack goods interface **263**. The container bot riding surface **266**RS may include rails **1200**S (see FIG. **1**B) that extend from the container transport deck **130**DC in a manner similar to that of the picking aisles **130**A, so that autonomous guided container bots **110** traverse or move into and out of the breakpack modules **266** along the rails **1200**S, and the rails **1200**S defines a queue lane **130**QL (FIG. **2**C) for the autonomous guided container bots **110** at the breakpack goods interface **263**. It is noted that where the container bot riding surface **266**RS is formed by rails **1200**S the riding surface may include an undeterministic turn around area **1200**UTA (that is similar to the open undeterministic container transfer deck **130**DC) on which the autonomous guided container bots **110** turn to transition between different travel portions (e.g., inbound and outbound) of the breakpack goods autonomous transport travel loop **234**. As can be seen in FIG. **2**C, the container bot travel surface **266**RS of the breakpack module **266** forms a travel loop **233** around which the autonomous guided container bots **110** travel to respectively transport along the container bot travel surface **266**RS travel loop **233** a supply container (e.g., case unit, pickface, remainder container, etc.) between the storage locations **130**S and a breakpack operation station **140** (and/or vice versa), and a breakpack goods container (also referred to as a breakpack container) **264** between the breakpack goods interface **263** and the breakpack goods container storage location **130**SB or a lift **150**A (and/or vice versa). The travel loop **233** provides the container bot **110** with random access to any and each breakpack goods interface locations **263**L of the breakpack goods interface **263** along the bot travel surface **266**RS, where the breakpack goods interface locations **263**L form an asynchronous product distribution system.

[0070] The goods transfer deck **130**DG forms a goods autonomous transport travel loop **234** disposed at the storage level **130**L. The goods transfer deck **130**DG is separate and distinct from the travel loop **233** formed by the container bot travel surface **266**RS, and has the breakpack goods interface **263** coupling respective edges of the container autonomous transport travel loop **233** of the container transfer deck **130**DC and the breakpack goods autonomous transport travel loop **234** of the goods transfer deck **130**DG. The goods autonomous transport travel loop **234** formed by the goods transfer deck **130**DG is disposed on a deck surface **130**DGS of a deck (e.g., goods transfer deck **130**DG) at a respective storage level **130**L, and the breakpack goods autonomous transport travel loop(s) **234** of the goods transfer deck **130**DG is disposed on a different deck surface **130**DGS of the deck (e.g., goods transfer deck **130**DG), separate and distinct from the deck surface **130**BS of the container bot travel surface **266**RS (formed by the container transfer deck **130**DC and/or rails **1200**S) where the container autonomous transport travel loop **233** is disposed. The breakpack goods autonomous transport travel loop **234** formed by the goods transfer deck **130**DG (and hence the goods travel deck **130**DG) is disposed to confine at least one autonomous goods bot **262** to the respective storage level **130**L. The at least one autonomous guided goods bot **262** is arranged or otherwise configured for transporting, along the breakpack goods autonomous transport travel loop **234** formed by the goods transfer deck **130**DG, one or more breakpack goods BPG (e.g., a pack that is unpacked from the supply container in a pack level sort or a unit/each unpacked from a pack in a unit/each level sort) between the breakpack operation station **140** and the breakpack goods interface **263**. The container bot(s) **110** is also configured to autonomously pick and place the breakpack goods containers **264** at the breakpack goods interface **263** as described herein. The breakpack goods interface **263** may be substantially similar to one or more of the transfer stations TS and buffer stations BS described herein and include an undeterministic surface (similar to that of the rack storage spaces **130**S described herein) upon which breakpack goods containers **264** are placed so as to form an undeterministic interface between the goods transfer deck **130**DG and the container transfer deck **130**DC.

[0071] The goods transfer deck **130**DG may facilitate a decanting process where goods are picked from one container (such as a supply container **265** or any other suitable standardized container **265**S) at the breakpack operation station **140** and consolidated with goods (generally of the same type) in another (e.g., outbound as noted below) supply container **265** or standardized container **265**S at the breakpack goods interface **263**, where the other supply container **265** or standardized container **265**S is returned to storage. Generally, supply containers **265** inbound to the breakpack modules **266** are picked until empty but only some (not all) of the goods from the inbound supply container may be decanted. Here, what may be referred to as outbound (i.e., outbound from the breakpack modules **266**) supply containers **265** or standardized containers **265**S (such as totes, trays, etc.) may be placed on the breakpack goods interface **263** by the container bot(s) **110** in a manner similar to that described herein for the breakpack goods containers **264** to facilitate the decanting process. In the decanting process, goods are removed from a supply container **265** (which may be an original product/good(s) case packaging) at the breakpack operation station **140** and consolidated into the outbound supply container(s) **265** or standardized container **265**S (e.g., having the same type of goods as those being removed at the breakpack operation station **140**) located on the breakpack goods interface **263**. Consolidation of goods having the same type from multiple supply containers **265** into a lesser number of supply containers **265** (and then returned to storage by the container bot(s) **110**) may increase the storage density of the automated storage and retrieval system **100** as the supply containers **265** stored in the storage racks can be maintained in a substantially "full" state (rather than having multiple containers that are less than full with the same type of goods therein. In some aspects, the decanted goods (in the standardized container or outbound supply container) are output from the storage and retrieval system **100** via the lifts **150** to be palletized as part of a pallet load (such as at output station **160**UT) or to be shipped individually (such as at output station **160**EC).

[0072] The autonomous guided goods bots **262** may be any suitable type of autonomously guided bot with a payload configured for holding breakpack goods, not product containers (e.g., case units, pickfaces, etc.). Each of the autonomous guided goods bots **262** has a payload hold configured dissimilar from a payload hold of the container bot **110**. The autonomous guided goods bots **262** are configured to autonomously travel unconstrained along and across the breakpack goods autonomous transport travel loop(s) **234** formed by the goods deck **130**DG and any suitable travel speeds, which may be the same as, greater than, or less than those travel speeds noted above with respect to the autonomous guided container bots **110**. The autonomous guided goods bots **262** are configured so as to automatically unload one or more breakpack goods BPG (retrieved from the breakpack operation station **140**) from the autonomous guided goods bot **262** to breakpack goods containers **264** at the breakpack goods interface **263**. Suitable examples of autonomous guided goods bots **262** include those produced by Symbotic of Wilmington, Massachusetts (United States), see for example, U.S. patent application Ser. No. 17/657,705 filed on Apr. 1, 2022 (Published as US PG Pub 2022/0289479), U.S. Provisional Patent Application No. 63/452,735 filed Mar. 17, 2023; and those produced by Tompkins International of Raleigh, North Carolina (United States), see for example, U.S. Pat. No. 10,248,112 issued on Apr. 2, 2019. The breakpack goods autonomous transport travel loop(s) **234** formed by the goods deck **130**DG has multiple travel lanes (see FIG. **2**C) for travel of the autonomous guided goods bots **262** along the breakpack goods autonomous transport travel loop(s) **234** (see, e.g., travel loops **234**A-**234**E) formed by the goods deck **130**DG. As noted herein, three travel lanes are illustrated for exemplary purposes only and in other aspects there may be more or less than three travel lanes. At least one of the multiple travel lanes is a passing lane for the autonomous guided goods bot **262** travel passing an obstruction on another of the multiple travel lanes in a manner similar to that described herein with respect to the multiple travel lanes of the container transfer deck **130**DC. The breakpack goods autonomous transport travel loop(s) **234** provide the autonomous guided goods bots **262** with random access to any and each of the breakpack goods interface locations **263**L of the breakpack goods interface **263**. In other aspects, the breakpack goods autonomous transport travel loop(s) **234** provide the autonomous guided goods bots **262** access to the belt sorter BST where the belt sorter BST sorts (and may be configured as a sorting buffer) the breakpack goods to the breakpack goods interface **263**. Here the belt sorter BST operates as an interface between the autonomous guided goods bots **262** and autonomous guided container bots **110**.

[0073] One or more portions of the goods transfer deck **130**DG (such as adjacent the breakpack goods interface locations **263**L) can be reserved to provide an exit (or off) ramp or entrance (or on) ramp from or to a travel loop travel **234**A-**234**E to effect a transfer of breakpack goods BPG to or from the breakpack goods container(s) **264** (or supply containers **265**, **265**S) at the breakpack goods interface locations **263**L. Exit ramps (referred to herein as ramps **222**, **222**C, **222**R) will be described herein but it should be understood that the entrance ramps are substantially opposite in direction to the exit ramps **222**, **222**C, **222**R (e.g., provide access to rather than access from a travel loop). One or more ramps **222**, **222**C, **333**R are provided depending on, for example, bot **110** kinematics (velocity, direction, etc.) and location(s) of (destination) breakpack goods interface locations **263**L (e.g., near corners of the goods transfer deck **130**DG, away from the corners of the goods transfer deck **130**DG, etc.) being accessed by the autonomous guided goods bots **262**. For exemplary purposes only, ramp **222** is a generic depiction of an on/off ramp that may be located anywhere on the goods transfer deck **130**DG and have any suitable length. Ramp **222**C is located in a corner of the goods transfer deck **130**DG. Ramp **222**R is a "rolling" ramp that moves to follow a path of an autonomous guided goods bot **262** traveling along the ramp **222**R,

[0074] The ramps **222**, **222**C, **222**R (both on and off ramps) may be "closed" temporarily from general access by the autonomous guided goods bots **262** (e.g., only predetermined autonomous guided goods bots delivering breakpack goods to and from the breakpack goods interface locations **263**L within the areas designated by the ramps **222**, **222**C, **222**R have access to respective on and off ramps). Generally the ramps **222**, **222**C, **222**R provide passage to and from a passing lane to a destination breakpack goods interface location **263**L. Each ramp **222**, **222**C, **222**R may be bidirectional (such as where a goods bot **2662** enters the ramp and travels in one direction along the ramp to pick or place a breakpack good BPG and then travels in the opposite direction along the ramp to exit from the ramp). The ramp may be a "counter-flow ramp" where travel along a ramp **222**, **222**C, **222**R is in a generally opposing direction to a travel direction around one or more of the travel loop(s) **234** (e.g., an autonomous guided goods bot **262** exits the travel loop and travels in the generally opposing direction along the ramp **222**, **222**C, **222**R). Where the ramp **222**, **222**C, **222**R is an off ramp, the ramp **222**, **222**C, **222**R may terminate at the destination breakpack goods interface location **263**L. Similarly, where the ramp **222**, **222**C, **222**R is an on ramp, the ramp **222**, **222**C, **222**R may begin at the destination breakpack goods interface location **263**L. As noted above, the ramps **222**, **222**C, **222**R may be located anywhere on the goods transfer deck **130**DG such that ramp entry locations vary in what may be referred to as a parking lane (e.g., a lane or a portion of a travel loop in which the goods bot stops to pick or place breakpack goods BPG) based on one or more of bot kinematics and locations of available breakpack goods interface locations **263**L. It is noted that while the turns of the autonomous guided goods bots **262** to and from the ramps **222**, **222**C, **222**R are illustrated as being substantially 90° turns, in other aspects, the turns may have an "S" shape similar to that described in U.S. patent application Ser. No. 16/144,668 filed on Sep. 27, 2018 and titled "Storage and Retrieval System", the disclosure of which is incorporated herein by reference in its entirety.

[0075] The ramps **222**, **222**C, **222**R are dynamically generated and may be dynamically effected (e.g., a "rolling" ramp, such as ramp **222**R) so that the ramp "rolls" in a progressive fashion with an initial ramp length generated from goods bot entry with adequate clearance for goods bot collision avoidance. In one or more aspects, the ramp **222**, **222**C **222**R is initiated (at bot entry) given that the ramp to the destination breakpack goods interface location **263**L is "blocked" (or otherwise obstructed) by an active autonomous guided goods bot **262**/active breakpack goods interface location **263**L but the blockage is expected to clear before the autonomous guided goods bot **262** traveling along the ramp reaches the blockage. If the blockage to the ramp **222**, **222**C, **222**R clears, the ramp **222**, **222**C, **222**R may be extended to the destination breakpack goods interface location **263**L; however, if the blockage does not clear the autonomous guided goods bot **262** travelling along the ramp **222**, **222**C, **222**R may be redirected to, for example, a passing lane and a new ramp is calculated/determined so that the autonomous guided goods bot **262** can place breakpack goods BPG at the destination breakpack goods interface location **263**L or another destination breakpack goods interface location **263**L.

[0076] Referring also to FIG. **2**D, the breakpack operation station **140** is configured so that one or more breakpack goods BPG are unpacked from supply container(s) **265** at the breakpack operation station **140**, and at least one autonomous guided goods bot **262** is configured so as to be loaded with the one or more breakpack goods BPG at the breakpack operation station **140**. An operator at the breakpack operation station **140** may place the breakpack goods BPG to the at least one autonomous guided goods bot **262** for transfer to the breakpack goods interface **263**. Referring to FIGS. **1** and **2**A, a belt sorter BST may be disposed between the breakpack operation station **140** and the autonomous guided goods bots **262** and forms an interface therebetween. Here the operator at the breakpack operation station places the breakpack goods BPG to the belt sorter BST where the belt sorter BST sorts (and in some aspects operates as a sorting buffer) the breakpack goods BPG to the autonomous guided goods bots **262**. The breakpack operation station **140** includes any suitable supply container **265** support surface **140**S. The support surface **140**S may be an undeterministic surface substantially similar to that of the storage shelves described herein and include slats **1210**S that form the support surface **140**S. The support surface **140**S may be an undeterministic roller conveyor (powered or unpowered), having rollers **140**RL with an arrangement similar to rollers **110**RL (see FIGS. **4**A and **4**B) of the container bot **110** described herein so that tines **273**A-**273**E of the pick head **270** of the container bot **110** (FIGS. **4**A and **4**B) are interdigitated with the rollers of the roller conveyor for placing (or picking) supply containers **265** to (or from) the support surface **140**S. Here, the container bot **110** is configured to autonomously transfer the supply container(s) **265** from the container bot **110** to the breakpack operation station **140** (such as to the support surface **140**S) in the manner described herein. Referring to FIGS. **1** and **2**A, the autonomous guided container bots **110** may deliver the supply containers **265** to a belt sorter BST configured as an interface between the autonomous guided container bots **110** and the breakpack operation station **140**. Here, the autonomous guided container bots **110** place the supply containers **265** to the belt sorter BST and the belt sorter BST sorts the supply containers **265** (and in some aspects operates as a sorting buffer) to the support surface **140**S of the breakpack operation station **140**. The support surface **140**S may be configured so that as the supply containers **265** are placed by the container bot **110** or belt sorter BST the supply containers **265** move along the support surface **140**S towards an operator **141** (e.g., a human operator or any suitable robotic operator (e.g., articulated arm, gantry, etc.)) for picking of breakpack goods BPG from the supply containers **265** and placement of the picked breakpack goods to autonomous guided goods bots **262** or to one or more of standardized containers **265**S (such as totes, trays, etc.) and breakpack goods containers **264** located at an operator staging area **140**A in any suitable manner to effect one or more of a pack level sortation of goods or a unit/each level sortation of goods. The supply containers **265** may be moved along the support surface **140**S to a respective operator staging area **140**A where the operator **141** picks the breakpack goods BPG from the supply containers **265** for placement in an autonomous guided goods bot **262** or in another container **265**S, **264**. The operator staging area **140**A may be contiguous with and/or formed by the support surface **140**S. As described herein, supply cases **265** with remaining goods therein after breakpack is performed may be picked by the autonomous guided container bots **110** from the support surface **140**S or staging area **140**A and returned to storage or to a lift **150**. Empty supply containers **265** may be removed from the support surface **140**S or staging area **140**A by the operator **141** and stored at the breakpack operation station **140** for later removal in any suitable manner. In one or more aspects, autonomous guided container bots **110** may transport empty containers from the storage and retrieval system via the lifts **150**. The breakpack operation station **140** may include any suitable refuse removal system **223** for removing refuse (or trash, e.g., shrink wrapping, packaging, boxes, etc.) from the storage and retrieval system. The refuse removal system **223** may include one or more of chutes, conveyors, lifts, or any other suitable transport configured to move refuse to a predetermined location; while in other aspects, the refuse may be placed in containers and removed from the storage and retrieval system by the autonomous guided container bots **110** via the lifts **150**. As can be seen in FIGS. **2**C and **13**, the breakpack goods transfer deck **130**DG joins the breakpack operation station **140** and the container transfer deck **130**DC at a separate location (e.g., at the breakpack goods interface locations **263**L) from each access of the container transfer deck **130**DC to the breakpack operation station **140** (e.g., at the common support surface **140**S) for the container bot **110**.

[0077] The autonomous guided container bots **110** may be any suitable independently operable autonomous transport vehicles that carry and transfer case units along the X and Y throughput axes throughout the storage and retrieval system **100**. The autonomous guided container bots **110** may be automated, independent (e.g. free riding) autonomous transport vehicles. Suitable examples of bots can be found in, for exemplary purposes only, U.S. Pat. No. 10,822,168 issued on Nov. 3, 2020; U.S. Pat. No. 8,425,173 issued on Apr. 23, 2013; U.S. Pat. No. 9,561,905 issued on Feb. 7, 2017; U.S. Pat. No. 8,965,619 issued on Feb. 24, 2015; U.S. Pat. No. 8,696,010 issued on Apr. 15, 2015; U.S. Pat. No. 9,187,244 issued on Nov. 17, 2015; U.S. Pat. No. 11,078,017 issued on Aug. 3, 2021; U.S. Pat. No. 9,499,338 issued on Nov. 22, 2016; U.S. Pat. No. 10,894,663 issued on Jan. 19, 2021; and U.S. Pat. No. 9,850,079 issued on Dec. 26, 2017, the disclosures of which are incorporated by reference herein in their entireties. The autonomous guided container bots **110** (described in greater detail below) may be configured to place case units, such as the above described retail merchandise, into picking stock in the one or more levels of the storage structure **130** and then selectively retrieve ordered case units. The throughput axes X and Y (e.g. pickface transport axes) of the storage array may be defined by the picking aisles **130**A, at least one container transfer deck **130**DC, the container bot **110** and the extendable end effector (as described herein) of the container bot **110** (and in other aspects the extendable end effector of the lifts **150** also, at least in part, defines the Y throughput axis).

[0078] The pickfaces (which may include supply containers **265**) are transported between an inbound section of the storage and retrieval system **100**, where pickfaces inbound to the array are generated (such as, for example, input station **160**IN) and a load fill section of the storage and retrieval system **100** (such as for example, output station **160**UT or output station **160**EC), where outbound pickfaces from the array are arranged to fill a load in accordance with a predetermined load fill order sequence or an individual fulfillment order(s) in accordance with a predetermined individual fulfillment order sequence. Pickfaces (e.g., of supply containers **265**) may be transported between the storage spaces **130**S and a load fill section of the storage and retrieval system **100** (such as for example, output station **160**UT or output station **160**EC) to fill a load in accordance with a predetermined load fill order sequence or an individual fulfillment order(s) in accordance with a predetermined individual fulfillment order sequence. Breakpack goods container(s) **264** (which may be multiple breakpack goods containers may be arranged in and transported as a pickface) may be transported between the storage spaces **130**S and the load fill section and/or between the breakpack goods interface **263** of the breakpack module(s) **266** and the load fill section of the storage and retrieval system **100** (such as for example, output station **160**UT or output station **160**EC) to fill a load in accordance with a predetermined load fill order sequence or an individual fulfillment order(s) in accordance with a predetermined individual fulfillment order sequence.

[0079] Referring also to FIGS. **1**A, **1**B, **1**C, and **2**A the rack module array RMA (see FIGS. **1**C and **2**A) of the storage structure **130** includes vertical support members **1212** and horizontal support members **1200** (see FIGS. **1**A and **1**B) that define the high density automated storage array as will be described in greater detail below. Rails **1200**S may be mounted to one or more of the vertical and horizontal support members **1212**, **1200** in, for example, picking aisles **130**A and be configured so that the autonomous guided container bots **110** ride along the rails **1200**S through the picking aisles **130**A. At least one side of at least one of the picking aisles **130**A of at least one storage level **130**L may have one or more storage shelves (e.g. formed by rails **1210**, **1200** and slats **1210**S).

[0080] Autonomous guided container bots **110** traversing a picking aisle **130**A, at a corresponding storage level **130**L, have access (e.g. for picking and placing case units and/or breakpack goods containers) to each storage space **130**S that is available on each shelf, where each shelf (which shelves may be disposed on one or more storage levels located between adjacent vertically stacked storage levels **130**L on one or more side(s) PAS**1**, PAS**2** of the picking aisle **130**A). Each storage space **130**S of the one or more storage shelf levels is accessible by the container bot **110** from the rails **1200** (e.g. from a common picking aisle deck **1200**S that corresponds with a container transfer deck **130**DC on a respective storage level **130**L).

[0081] Referring again to FIG. **2**A each container transfer deck **130**DC or storage level **130**L may include one or more lift pickface interface/handoff stations TS (referred to herein as interface stations TS) where case unit(s) (e.g. individual case units, pickfaces, supply containers, etc.), totes and/or breakpack goods containers **264** are transferred between the lift load handling devices LHD and autonomous guided container bots **110** on the container transfer deck **130**DC. The interface stations TS are located at a side of the container transfer deck **130**DC opposite the picking aisles **130**A and rack modules RM, so that the container transfer deck **130**DC is interposed between the picking aisles and each interface station TS. As noted above, each container bot **110** on each picking level **130**L has access (via a respective container transfer deck **130**DC) to each storage location **130**S, each picking aisle **130**A, and each lift **150** on the respective storage level **130**L, as such each container bot **110** also has access to each interface station TS on the respective level **130**L. The interface stations TS may be offset from high-speed bot travel paths HSTP along the container transfer deck **130**DC so that container bot **110** access to the interface stations TS is undeterministic to bot speed on the high speed travel paths HSTP. As such, each container bot **110** can move a case unit(s) (e.g. individual case units, pickfaces (built by the bot), supply containers, etc.), totes and/or breakpack goods containers **264** from every interface station TS to every storage space **130**S corresponding to the deck level **130**L and vice versa.

[0082] The interface stations TS may be configured for a passive transfer (e.g. handoff) of case units (e.g. individual case units, pickfaces, supply containers, etc.), totes and/or breakpack goods containers **264** between the container bot **110** and the load handing devices LHD of the lifts **150** (e.g. the interface stations TS have no moving parts for transporting the case units) which will be described in greater detail below. For example, also referring to FIG. **2**B the interface stations TS and/or buffer stations BS include one or more stacked levels TL**1**, TL**2** of transfer rack shelves RTS (e.g. the container bot **110** being configured to access each of the one or more stacked levels TL**1**, TL**2**) which in one aspect are substantially similar to the storage shelves described above (e.g. each being formed by rails **1210**, **1200** and slats **1210**S) such that container bot **110** handoff (e.g. pick and place) occurs in a passive manner substantially similar to that between the container bot **110** and the storage spaces **130**S (as described herein) where the case units or totes are transferred to and from the shelves. The buffer stations BS on one or more of the stacked levels TL**1**, TL**2** may serve as a handoff/interface station with respect to the load handling device LHD of the lift **150**. Load handling device LHD (or lift) may handoff (e.g. pick and place) of case units (e.g. individual case units, pickfaces, supply containers, etc.), totes and/or breakpack goods containers **264** to the stacked rack shelves RTS (and/or the single level rack shelves) occurs in a passive manner substantially similar to that between the container bot **110** and the storage spaces **130**S (as described herein) where the case units, totes and/or breakpack goods containers **264** are transferred to and from the shelves. The shelves may include any suitable transfer arms for picking and placing case units, totes and/or breakpack goods containers **264** from one or more of the container bot **110** and load handling device LHD of the lift **150**. Suitable examples of an interface station with an active transfer arm are described in, for example, U.S. Pat. No. 9,694,975 issued on Jul. 4, 2017, the disclosure of which is incorporated by reference herein in its entirety.

[0083] The location of the container bot **110** relative to the interface stations TS may occur in a manner substantially similar to bot location relative to the storage spaces **130**S. For example, location of the container bot **110** relative to the storage spaces **130**S and the interface stations TS may occur in a manner substantially similar to that described in U.S. Pat. No. 9,008,884 issued on Apr. 14, 2015 and U.S. Pat. No. 8,954,188 issued on Feb. 10, 2015, the disclosures of which are incorporated herein by reference in their entireties. For example, referring to FIGS. **1** and **1**A, the container bot **110** includes one or more sensors **110**S that detect the slats **1210**S or a locating feature **130**F (such as an aperture, reflective surface, RFID tag, etc.) disposed on/in the rail **1200**. The Slats and/or locating features **130**F are arranged so as to identify a location of the container bot **110** within the storage and retrieval system, relative to e.g. the storages spaces and/or interface stations TS. The container bot **110** may include a controller **110**C that, for example, counts the slats **1210**S to at least in part determine a location of the container bot **110** within the storage and retrieval system **100**. The location features **130**F may be arranged so as to form an absolute or incremental encoder which when detected by the container bot **110** provides for a container bot **110** location determination within the storage and retrieval system **100**. The sensor **110**S may be configured to localize the container bot **110** through any suitable image processing of images of case units disposed on the storage selves.

[0084] One or more peripheral buffer/handoff stations BS (substantially similar to the interface stations TS and referred to herein as buffer stations BS) may be located at the side of the container transfer deck **130**DC opposite the picking aisles **130**A and rack modules RM, so that the container transfer deck **130**DC is interposed between the picking aisles and each buffer station BS. The peripheral buffer stations BS may be interspersed between or, as shown in FIGS. **2**A and **2**B, otherwise in line with the interface stations TS. The peripheral buffer stations BS may be formed by rails **1210**, **1200** and slats **1210**S and may be a continuation of (but a separate section of) the interface stations TS (e.g. the interface stations and the peripheral buffer stations are formed by common rails **1210**, **1200**). As such, the peripheral buffer stations BS may include one or more stacked levels TL**1**, TL**2** of transfer rack shelves RTS as described above with respect to the interface stations TS, although the buffer stations may include a single level of transfer rack shelves. The peripheral buffer stations BS define buffers where case units/totes/breakpack goods containers and/or pickfaces are temporarily stored when being transferred from one container bot **110** to another different container bot **110** on the same storage level **130**L as will be described in greater detail below. The peripheral buffer stations may be located at any suitable location of the storage and retrieval system including within the picking aisles **130**A and anywhere along the container transfer deck **130**DC.

[0085] Still referring to FIGS. **2**A and **2**B, at least the interface stations TS may be located on an extension portion or pier **130**BW (also referred to as a driveway) that extends from the container transfer deck **130**DC, although a length of the interface stations TS may be arranged and extend along the container transfer deck. The pier **130**BW may be similar to the picking aisles where the container bot **110** travels along rails **1200**S affixed to horizontal support members **1200** (in a manner substantially similar to that described above). The travel surface of the pier **130**BW may be substantially similar to that of the container transfer deck **130**DC. Each pier **130**BW is located at the side of the container transfer deck **130**DC, such as a side that is opposite the picking aisles **130**A and rack modules RM, so that the container transfer deck **130**DC is interposed between the picking aisles and each pier **130**BW. The pier(s) **130**BW extends from the transfer deck at a non-zero angle relative to at least a portion of the high speed bot transport path HSTP. The pier(s) **130**BW may extend from any suitable portion of the container transfer deck **130**DC including the ends **130**BE**1**, **130**BE**2** of the container transfer deck **130**DCD. Peripheral buffer stations BSD (substantially similar to peripheral buffers stations BS described above) may be located at least along a portion of the pier **130**BW.

[0086] As described above, and referring to FIG. **3**, the autonomous guided container bots **110** and the autonomous guided goods bots **262** may be non-holonomic. For example, referring to the container bot **110** for exemplary purposes (noting the autonomous guided goods bots **262** have a similar non-holonomic configuration), the goods bot **110** includes two drive wheels **202**A, **202**B located on opposite sides of the bot **110** at end **110**E**1** (e.g. first longitudinal end) of the bot **110** for supporting the bot **110** on a suitable drive surface however, in other aspects any suitable number of drive wheels are provided on the bot **110**. Each drive wheel **202**A, **202**B may be coupled substantially directly to a respective motor **202**MA, **202**MB such that the drive wheel **202**A, **202**B is coupled to an output of the motor **202**MA, **202**MB without a speed reduction unit disposed therebetween (e.g. so that each motor **202**MA, **202**MB and the respective drive wheel **202**A, **202**B form a reductionless drive). Each drive wheel **202**A, **202**B may be independently controlled so that the bot **110** may be steered through a differential rotation of the drive wheels **202**A, **202**B (e.g. differential torque steering), although the rotation of the drive wheels **202** may be coupled so as to rotate at substantially the same speed. Any suitable steering wheels **201** are mounted to the frame on opposite sides of the bot **110** at end **110**E**2** (e.g. second longitudinal end) of the bot **110** for supporting the bot **110** on the drive surface. The wheels **201** may be caster wheels that freely rotate allowing the bot **110** to pivot through differential rotation of the drive wheels **202** for non-holonomically changing a travel direction of the bot **110**. The wheels **201** may be steerable wheels, such as for example articulated wheel steering, that turn under control of, for example, a bot controller **110**C (which is configured to effect control of the bot **110** as described herein) for changing a travel direction of the bot **110**. The bot **110** may include any suitable wheel arrangement (e.g. a three wheel configuration, a four wheel configuration, etc.). The bot **110** may include one or more guide wheels **110**GW located at, for example, one or more corners of the frame **110**F. The guide wheels **110**GW may interface with the storage structure **130**, such as guide rails **1200** (FIG. **1**B) within the picking aisles **130**A, guide rails (not shown) on the transfer deck **130**B and/or at interface or transfer stations for interfacing with the lift modules **150** for guiding the bot **110** and/or positioning the bot **110** a predetermined distance from a location to/from which one or more case units are placed and/or picked up as described in, for example, U.S. Pat. No. 9,561,905 issued on Feb. 7, 2017, the disclosure of which is incorporated herein by reference in its entirety. Location of the bot **110** at the predetermined distance from a location to/from which one or more case units are placed and/or picked up may be effected in any suitable manner such as with sonar/sonic sensors, index or line following sensor, GPS sensors, inductive sensors, capacitive sensors, infrared sensors, computer vision sensors, etc. of the container bot **110** or any combination thereof.

[0087] As described herein, and referring to FIG. **4**, the container transfer deck **130**DC may have an undeterministic transport surface **130**BS on which the autonomous guided container bots **110** travel where the undeterministic transport surface **130**BS has more than one juxtaposed travel direction or lane HSTP (which correspond at least in part to a guidance feature of navigation array **3000** that is disposed on the undeterministic transport surface **130**BS) connecting the aisles **130**A, transfer deck **130**B, and the pier **130**BW to each other. While the navigation array is described herein with respect to the container transfer deck **130**DC, it is noted the breakpack goods transfer deck **130**DG may be similar to the container transfer deck **130**DC and include a respective navigation array **3000** having the features described herein with respect to the container transfer deck **130**DC. The juxtaposed travel lanes (the terms “travel lane” and “direction” may be used interchangeably herein) are juxtaposed along a common undeterministic transport surface **130**BS between opposing sides **130**BD**1**, **130**BD**2** (see also FIG. **2**A) of the transfer deck **130**B. For example, FIG. **4** illustrates longitudinal lanes such as an aisle side travel lane LONG**1**, a lift side travel lane LONG**3** and a pass through travel lane LONG**2**, but it should be understood that in other aspects more or fewer travel lanes are provided. Juxtaposed lateral travel lanes, such as lanes LAT**1**-LAT**7** may be arranged on the transfer deck **130**B to allow bot traverse to and from picking aisles **130**A, driveways **130**BW, transfer stations TS, buffer stations BS, and/or any other suitable location of the storage and retrieval system that is accessed through a path that is transverse to the longitudinal lanes.

[0088] As can be seen in FIG. **4** the navigation array **3000** is illustrated, for exemplary purposes, as a grid of linearly distributed features. The linearly distributed features LDF may include longitudinal features LONG**1**-LONG**3** and lateral features LAT**1**-LAT**7** where the longitudinal and lateral directions are relative to the transfer deck (e.g. the longitudinal features LONG**1**-LONG**3** define at least one of the travel lanes HSTP and the lateral features LAT**1**-LAT**7** define travel lanes HSTT that cross the travel lanes HSTP. The linearly distributed features LDF may allow for container bot **110** traverse between offset (parallel and/or crossing) travel lanes HSTP, HSTT in the storage structure **130** for entry into aisles **130**A, driveways **130**BW, buffer stations BS, transfer stations TS or any other suitable location at which the bot **110** performs an operation (transfer of cases, charging of the bot, bot induction into the storage structure, bot removal from the storage structure, etc.). The linearly distributed features LDF may allow for bot traverse between crossing travel lanes HSTP, HSTT for entry into aisles **130**A, piers **130**BW, buffer stations BS, transfer stations TS or any other suitable location at which the bot **110** performs an operation (transfer of cases, charging of the bot, bot induction into the storage structure, bot removal from the storage structure, etc.) where when the bot **110** detects a linear distributed feature the bot **110** establishes a location of the bot **110** during travel as will be described in greater detail below. One or more of the aisles **130**A and piers **130**BW may be “dead ends” in that there is an entry into the aisle **130**A or pier **130**BW from the container transfer deck **130**DC but there is no exit for the container bot from the aisle **130**A or pier **130**BW unless the container bot reverses direction and exits the aisle **130**A or pier **130**BW at the same point the bot entered the aisle **130**A or pier **130**BW (i.e., the aisle or pier has restricted access such that only a single point of ingress/egress to/from the aisle or pier exists).

[0089] The linearly distributed features LDF may connect the aisles **130**A to each other, cross the aisles **130**A, connect the aisles **130**A to one or more of the transfer stations TS, the buffer stations BS and the piers **130**BW or any combination thereof. One or more of the linearly distributed features LDF may be substantially aligned with one or more of the interface between the transfer deck **130**B and the aisles **130**A, and the interface between the transfer deck **130**B and the piers **130**BW. As noted above, at least a portion of the linearly distributed features LDF may be substantially aligned with one or more bot traverse paths **3010** along the transfer deck **130**B. It is noted that while the linearly distributed features LONG**1**-LONG**3**, LAT**1**-LAT**7** are illustrated as forming an orthogonal grid in other aspects the longitudinal features LONG**1**-LONG**3** and the lateral features LAT**1**-LAT**7** cross each other at any suitable angles. While three longitudinal features LONG**1**-LONG**3** (defining, for example, at least in part three travel lanes HSTP) and seven lateral features LAT**1**-LAT**7** (defining, for example, at least in part seven travel lanes HSTT) are described, the transfer deck **130**B may include any suitable number of longitudinal and lateral features LONG**1**-LONG**3**, LAT**1**-LAT**7** defining at least in part any suitable number of travel lanes oriented in any suitable directions relative to the transfer deck **130**B.

[0090] The linearly distributed features LDF may be formed of, for example, any suitable guide tapes, any suitable transfer deck **130**B features (grooves, apertures, channels, etc.), and edge of the transfer deck **130**B or any combination thereof. The linearly distributed features LDF may be uncoded (e.g. do not include identifying features such as for determining container bot **110** location) while in other aspects the linearly distributed features are coded (e.g. include or are formed of barcodes or other identifying indicia or features so as to provide for container bot **110** location determination). It is noted however, that the linearly distributed features may be placed at predetermined locations on the transfer deck **130**B to allow the bot to establish at least an estimated location of the container bot **110** while travelling at high speeds (as previously described) along the transfer deck **130**B. With reference to FIG. **3**, the spacing between juxtaposed linearly distributed features LDF is not dependent on container bot **110** dimensions or operational aspects, such as bot longitudinal wheelbase LONWB (where the container bot **110** has a longitudinal axis LX and a lateral axis LT), wheel track or lateral wheelbase LATWB, turn radius and bot width (in the lateral direction). The bot frame **110**F, longitudinal wheelbase LONWB and lateral wheelbase LATWB may define a predetermined aspect (such as for example, a ratio of length to width) that provides the non-holonomic container bot **110** with a minimum turn radius (and/or minimum turn radius footprint defined by the outermost corners of the frame turning at the minimum turn radius). The spacing between juxtaposed travel lanes LONG**1**-LONG**3**, LAT**1**-LAT**7** may be such that it allows passage of two autonomous guided container bots **110** (travelling linearly) side by side on each lane but is less than the minimum turn radius of the non-holonomic container bot **110** for a 90° pivot turn (pivoting at the drive wheels **202**—see FIG. **3**, such as at a pivot location/axis disposed between the drive wheels **202**A, **202**B)). As such the outer lanes (e.g. the aisle side travel lane and the lift side travel lane and/or corresponding lanes at the ends **130**BE**1**, **130**BE**2** of the container transfer deck **130**DC) are, in one aspect positioned close to the sides of the container transfer deck **130**DC (by an amount just sufficient to allow the container bots **110** to travel along the travel lane) but less than the spacing required for the container bot **110** to make a 90° pivot turn (pivoting at the drive wheels **202**—see FIG. **3**). The position of the picking aisles **130**A, the lift interfaces/transfer stations TS and buffers stations BS relative to the container transfer deck **130**DC and each other are decoupled from bot **110** turn considerations.

[0091] It is noted that for descriptive purposes only, the intersections between the linearly distributed features are referred to as nodes ND so that the transfer deck surface **130**BS and its associated features (e.g. the linearly disturbed features LDF) are represented as a grid (as described above) with an array of nodes. The nodes ND may be disposed at any suitable predetermined location on the longitudinal and/or lateral features LONG**1**-LONG**3**, LAT**1**-LAT**7** (such as at an intersection) of the linearly distributed features LDF that may correspond, for example, to a feature of the storage structure **130** and/or navigation array **3000** (e.g. at a terminus to a storage aisle **130**A, at a lift transfer station TS, an entry to a pier **130**BW, at a buffer station BS or at any other suitable location of the container transfer deck **130**DC). It should be understood that the concept of a node ND as used herein is to exemplify that the navigation array **3000** defines the linearly distributed features LDF which map out the container transfer deck **130**DC in two dimensions where the array of nodes ND on the deck are associated with the longitudinal and lateral features LONG**1**-LONG**3**, LAT**1**-LAT**7**. As will be described in greater detail below, waypoints WP**1**-WP**2** that lay along a container bot **110** travel path may be created at predetermined locations on the container transfer deck **130**DC where in some aspects one or more waypoints WP**1**-WP**4** may coincide with one or more nodes ND and, as with the nodes ND, may be positioned on linearly distributed features LDF defining a respective linear direction. One or more of the waypoints WP**1**-WP**4** may be located between nodes ND, be located offset from the nodes ND in any suitable direction, or be located offset from the linearly distributed features LDF in any suitable direction.

[0092] As described herein, the container bots **110** travel along time optimal paths and trajectories, examples of which are illustrated in FIG. **4** with respect to transport paths BTP-BTP**3** where each of the transport paths embody respective trajectories. The transport path BTP includes waypoints WP**1**, WP**2**, WP**3**, and WP**4**. The transport path BTP**2** includes waypoints WP**5** and WP**6**. The transport path BPT**3** includes waypoints WP**6** and WP**7**. The transport paths may include straight paths, arcuate paths, compound paths forming shapes such as “S” shape curves, or any other combination of straight and arcuate paths. The time optimal paths and trajectories may be generated in any suitable manner such as the manner described in U.S. Pat. No. 11,760,570 issued on Sep. 19, 2023 the disclosure of which is incorporated herein by reference in its entirety. The time-optimal trajectories **990**A-**990***n*, **991**A-**991***n*, **992**A-**992***n*, **993**A-**993***n*, **994**A-**994***n* and respective time optimal paths embodied thereby allow the bot **110** to smoothly transition, while travelling at the high speeds described herein, along a smooth curved (curve, curves, or curvature is used generally herein to refer to traverse path lines with concave/convex shape, through the techniques applied herein are equally applicable to path lines with flat curves that will be expressly identified as such) path defined by the corresponding trajectory between two navigation features arranged along a common direction (e.g. such as between one or more of the longitudinal navigation features LONG**1**-LONG**3** or between one or more of the lateral navigation features LAT**1**-LAT**7**) and/or between two navigation features that are arranged in different directions (e.g. such as between longitudinal navigation features LONG**1**-LONG**3** and lateral navigation features LAT**1**-LAT**7**, see FIG. **4**). The time-optimal trajectories **990**A-**990***n*, **991**A-**991***n*, **992**A-**992***n*, **993**A-**993***n*, **994**A-**994***n*, based on bot dynamic model(s), may be categorized or catalogued, in a table **6042**T (such as stored in a memory **110**M, **262**M of the bot controller **110**C, **262**C of a respective autonomous guided container bot **110** and/or autonomous guided goods bot **262**—see FIG. **4**A), in any suitable manner. The trajectories are according to their respective characteristics shown categorized according to linear length L, L**1**, . . . , Ln along the curved path defined by the corresponding trajectory where the length L, L**1**, . . . , Ln may correspond to a linear distance to be travelled by the bot **110** between positions P**1** (RX**1**, RY**1**) and P**2** (RX**2**, RY**2**) (see FIGS. **3** and **4**) defined within the bot **110** map coordinate system REF as specified in a bot motion command received from, for example, controller **120**.

[0093] Referring now to FIGS. **4**, **5**A, and **5**B, exemplary bot traverse paths of an exemplary container bot **110** are shown in accordance with the present disclosure. These transport paths may be for any suitable bot such as, for example, the container bots **110** and/or autonomous guided goods bots **262** described herein. The time-optimal trajectory or trajectories may be generated (in any suitable manner, such as described in U.S. Pat. No. 11,760,570, previously incorporated herein by reference in its entirety) for each of these exemplary traverse paths. It is noted that the term "traverse path" as used herein and depicted in the drawings refers to the physical traverse paths of the autonomous guided container bots **110** and/or autonomous guided goods bots **262** on/along the undeterministic surface **130**BS defined by the open undeterministic container transfer deck **130**DC (and similar undeterministic surface defined by the open and undeterministic goods deck **130**DG) or floor between locations thereon. It is noted that the traverse paths may include one or more segments (such as those described herein with respect to FIGS. **4** and **4**A) that are joined to each other to form the traverse path. The term "trajectory" of the traverse path includes kinematic properties of the respective autonomous guided container bot **110** and autonomous guided goods bot **262** such as, for example, acceleration, velocity, etc. moving along and defining the traverse path.

[0094] As described above, any suitable controller, such as controller **110**C, **262**C of the respective autonomous guided container bot **110** and autonomous guided goods bot **262**, may be configured as a bang-bang controller for generating time-optimal motions of the bot **110** using maximum power of the respective autonomous guided container bot **110** and autonomous guided goods bot **262** drive section. It is noted the present disclosure allows for the generation of otherwise predetermined autonomous guided bot **110**, **262** unparameterized autonomous guided bot **110**, **262** trajectories having motor torque (e.g. maximum torque/peak usable torque) and/or boundary constraints for, e.g., different autonomous guided bot **110**, **262** payload applications or any other velocity, acceleration, etc. constraints. The term unparameterized as used herein with respect to the generated trajectories means that the trajectory and traverse path characteristics are unconstrained as to the curve or shape of the trajectory (either with respect to time or in the position-velocity reference frame or space) such that a time-optimal trajectory shape is achieved within the noted constraints of available autonomous guided bot **110**, **262** maximum motor torque (e.g., the desired maximum usable torque for the maximum available current from the autonomous guided bot **110**, **262** power source, and other bot dynamic models (e.g., mass, moment of inertia, radius of drive wheels, drive wheel base, etc.) and initial and final inertial conditions). Trajectories can be generated for each of the traverse path segments such that optimal (shortest) move times (e.g. autonomous guided bot **110**, **262** traverse times between a starting point of the traverse path and an ending point of the traverse path) are achieved for given maximum drive torque constraints. Further, peak torque requirements for drive components, such as motors and/or gear boxes (if any), can be reduced (with or without shorter move times) leading to lower costs associated with the autonomous guided bot **110**, **262**, reduced size of the drive components, and/or increased life of the autonomous guided bot **110**, **262**.

[0095] As used herein, the term "smooth" or "smoothness" with respect to the generated trajectories refers to a continuous linear velocity along the curved traverse path over time. It is noted that a discontinuity in linear velocity is generally not practically achievable, given the autonomous guided bot **110**, **262** inertial and dynamic characteristics at high and medium speeds, and undesired.

[0096] The time-optimal trajectories may be categorized based on autonomous guided bot **110**, **262** dynamic model characteristics and/or other boundary conditions, such as a bot payload (e.g., empty or loaded bot), payload mass and/or size where more massive payloads and/or denser payloads (e.g., resulting in bot mass center eccentricity) may define larger radius/curved turns at the high speeds. The trajectories may be categorized based on one or more of a distance to be travelled by the autonomous guided bot **110**, **262** and a payload weight/mass and/or size/mass distribution or payload density.

[0097] As may be realized from FIGS. **4** and **4**A, there are different types of predetermined time-optimal dynamic model based trajectories that define different path curves the autonomous guided bot **110**, **262** can dynamically select by the bot controller **110**C, **262**C (or any other suitable controller such as control server **120** or warehouse management system **2500**) from, singly and/or in a dynamically selected combination applied consecutively or separated by a flat path line (see FIG. **4**A), to follow from an initial position (with an initial static and/or on the fly pose) to a final position (with a final desired/resultant static and/or on the fly pose). The trajectories may be characterized by simple curves having any suitable shape such as a straight line, semi-circles and hook or "J" shapes while in other aspects the trajectories are any suitable complex curves such as "S" shaped curves having multiple changes in direction or inflection points as described. For example, one type of time-optimal trajectory is a component time-optimal trajectory having curved portions and straight portions where the curved portions are compound curves or turns with one or more of a variable or constant turn radius/rate, constant or changing turn directions and simple turns. As another example of a time-optimal trajectory type, the characterizing path shapes of the time optimal trajectories are a simple turn (turning in one direction during the move) having one or more of a variable turn radius/rate (e.g., the trajectory of paths **991**A-**991***n* and **992**A-**992***n* illustrated in FIG. **4**A) or constant turn radius/rate (e.g., the trajectory of paths **990**A-**990***n* illustrated in FIG. **4**A) in a single direction. The different component time-optimal trajectories (e.g. simple and compound) may be joined to each other, end to end, to form a transport path BTP (e.g., as shown in FIG. **4**, having a time-optimal trajectory characterized by an "S" shape (e.g., such as the trajectory of paths **993**A-**993***n* shown in FIG. **4**A) combined with a time-optimal trajectory characterized by a simple curve/turn; although a flat curve path trajectory (e.g., such as the trajectory of paths **994**A-**994***n* shown in FIG. **4**A) may be joining the bending curve path trajectory) followed by the autonomous guided bot **110**, **262**.
[0098] Referring now to FIGS. **4**, **4**A, **5**A and **5**B, conventionally non-holonomic autonomous guided bot **110** traverse across the container transfer deck **130**DC or a goods deck **130**DG is facilitated by, for example (see FIG. **5**A) line following where the guidelines are arranged in a grid pattern such that bot (autonomous guided vehicle) turns are limited to 90° turns where the 90° turns are made at nodes ND (e.g. crossing points) of the guidelines. As may be realized from the above, the high speed navigation of the autonomous guided bots **110**, **262** described herein allows substantially unrestricted non-holonomic autonomous guided bot **110**, **262** traverse (see FIG. **5**B) along the undeterministic surface of a respective one of the open undeterministic container transfer deck **130**DC and the open undeterministic breakpack goods transfer deck **130**DG such the bot **110**, **262** can travel from node to node without being restricted to making 90° turns at the nodes ND, thereby decreasing autonomous guided bot **110**, **262** transfer/travel times across the respective container transfer deck **130**DC and breakpack goods transfer deck **130**DG. This allows the autonomous guided bot **110**, **262** to travel at the speeds described herein while navigating the undeterministic surface of the respective container transfer deck **130**DC and breakpack goods transfer deck **130**DG and while making turns into the picking aisles **130**A and/or lift interface areas such as the driveways **130**BW or buffer/transfer stations located along a side of the container transfer deck **130**DC (such as shown in FIGS. **2**A-**2**C) or to breakpack stations **140** and breakpack goods interface location(s) **263**L of a breakpack goods interface **263** of the breakpack goods transfer deck **130**DG.
[0099] Referring to FIGS. **1** and **6**A, as described herein, autonomous guided bot **110**, **262** route planning may be effected by one or more of the controller **120** and warehouse management system **2500** (or any other suitable controller of the storage and retrieval system **100**). Bot route planning will be described with respect to controller **120** for exemplary purposes only, where bot route planning may be effected in a similar manner when performed by the resolver **2500**PR of the bot route planner **2500**P of the warehouse management system **2500**.
[0100] As described herein, the controller **120** includes the resolver **120**PR that is configured with

the bot route planner **120**P. The bot route planner **120**P is configured to obtain or otherwise collect unplanned route legs URL and effect planning of the bot routes based on bot priority. Here, the resolver **120**PR divides or otherwise sequences the unplanned route legs URL into batches BRL. An example for determining a of batch route legs will now be described, where the bath of route legs BRL meets the predetermined maximum number N of route legs.

[0101] In determining a batch, the resolver **120**PR initializes a batch B. If this batch B is the first batch, the batch B is initialized (FIG. **10**, Block **1000**) to be the route legs that are recursively safety-threatened by active route legs; otherwise, the batch BRL is initialized to be an empty set.

[0102] With respect to batching of route legs, decisions made by the resolver **120**PR for batching or otherwise grouping route legs are defined. Given a batch size limit (i.e., the predetermined maximum number N of route legs), a batching decision BRL of a set of route legs is a sequence BRLS of batches BRL**1**-BRLn that exactly cover all the route legs, and each batch BRL**1**-BRLn does not exceed the predetermined maximum number N of route legs in a batch. A partial batching decision PB is also defined as a batching decision of a subset of given route legs. All the batching decisions that can be extended from a partial batching decision PB for un selected route leg set R is defined as a Full Decision Set Extend (PB, R).

[0103] The batching decision obtained by adding route leg r into the current open batch of partial batch decision is defined as PB+r; and the batching decision obtained by adding route leg r into the next batch of partial batch decision is defined as PB++r.

[0104] The quality of a batching decision B is the objective value of the planning solution obtained by planning route legs batch by batch by following this batching decision. The quality of batching decision BRL is denoted as q (BRL), where objective values are maximized).

[0105] The quality of a partial batching decision PB is the maximum quality of Extend (PB, R). The it quality of a partial batching decision PB is denoted as q (PB, R)=max q (BRL) over BRL in Extend (PB, R).

[0106] The quality of batching decisions may be compared by where the priority-based divide and search autonomous transport vehicle path planning process PBDS is considered an optimal single batch scheduler, and the objective value of planning all route legs as one batch is Q*. This is the best objective value of any solution and the batching decision quality bound. Here, every batching decision's objective value is not smaller than Q*. For two batching decisions B**1** and B**2**, we say B**1** is better than B**2** if q (B**1**)>q (B**2**). A good batching decision is expected to lead to a solution as close as possible to the solution obtained by directly planning all the route legs as one batch. For example, assume there are four route legs A, B, C, D and the minimized objective value is 10 by planning all of the four route legs as one batch. With a batch size limit (i.e., predetermined maximum number N of route legs in a batch) of two, if {(A, B), (C, D)} has quality 12, and {(A, C), (B, D)} has quality 15, we say the latter batching decision is better.

[0107] One example of approximating a batch determination is based on a partial batching decision PB, and unplanned route legs (r1, r2, . . . , rn). Here, deciding which route leg to select next to further extend PB, includes selecting the partial batching decision with the best quality from (PB+r1, PB+r2, . . . , PB+rn). However, it is hard to evaluate the quality of a partial batching decision PB+r, where the quality of all the decisions in Extend (PB+r, R−r) is evaluated. The number of Extend (PB+r, R−r) can grow exponentially to the number of unselected legs.

[0108] In accordance with the present disclosure, an approximation method is employed by the resolver **120**PR, **2500**PR to estimate impact of prioritizing a route leg r by evaluating the route leg's impact only: As g(PB, R) is a constant for every route leg selection, we evaluate g(PB, R)−g(PB+r, R−r); the effect of other unselected legs is ignored, which leads to evaluating g(PB+r)−g(PB, [r]); and as g(PB, [r])=max(g(PB+r), g(PB++r)), g(PB+r)−g(PB, [r])=max(0, g(PB+r)−g(PB++r).

[0109] This value g(PB+r)−g(PB++r) can be explained as follows: g(PB+r) is the batching decision quality of putting this leg in the current open batch; g(PB++r) is the batching decision quality of putting this leg in the next batch; and g(PB+r)−g(PB++r) is the objective value difference made by

de-prioritizing leg r.
[0110] Larger the value for g(PB+r)−g(PB++r), the more urgent to put r into the current batch. This leads to a rule of route leg batching selection: pick the route leg that impacts the objective value most when being deferred into the next batch.
[0111] However, the above rule may be hard to apply in batching because in determining g(PB+r)−g(PB++r) the corresponding two objective values are determined and the difference between the two objection values are calculated. For example, putting r in the current batch or next batch may not only impact route leg r's total time to reach the destination. Here, other route legs may not get impacted a lot and another example of approximating a batch determination is based on only evaluating the impact of safety, reachability, and delay of the route leg r. The evaluation of only evaluating the impact of safety, reachability, and delay of the route leg r leads to a fundamental rule of route leg batching selection: pick the route leg that is most impacted by being deferred into the next batch.
[0112] The resolver **120**PR evaluates a route leg deferral penalty (as described in greater detail herein) and initializes deferral penalties for all route legs (FIG. **10**, Block **1010**) given the threat graph(s) (see FIG. **7**). For example, the resolver **120**PR finds the route leg r with the largest deferral penalty (FIG. **10**, Block **1015**). The resolver **120**PR finds a set of route legs R that includes the route leg r and all the route legs that are recursively safety-threatened by route legr (FIG. **10**, Block **1020**). The resolver **120**PR adds the set of route legs R to the batch B and returns the batch B (FIG. **10**, Blocks **1025**, **1026**, and **1027**) based on the following three conditions: add and return the batch B if the batch size is the predetermined maximum number N of route legs (FIG. **10**, Block **1021**); if adding the set of route legs R to the batch B will exceed the predetermined maximum number N of route legs (FIG. **10**, Block **1022**): return the batch B if the batch B is non-empty or return the set of route legs R if the batch B is empty (FIG. **10**, Block **1023**); and adds but does not return the batch B if the new batch size (e.g., the sum of the set of route legs R and the batch B is smaller than the predetermined maximum number N of route legs (FIG. **10**, Block **1030**). The resolver **120**PR updates the deferral penalties of route legs that are delayed or reachability is threatened by the route legs in the set of route legs R (FIG. **10**, Block **1040**). Blocks **1015-1040** of FIG. **10** may be repeated until there are no more route legs to include in a batch B.
[0113] All of the route legs that are recursively safety-threatened by one or multiple route legs are added to the batch B. This is implemented as: initializing a route leg set R with the given route legs; and repeating the following (until no route leg is added) add all the route legs that are safety-threatened or same-bot-threatened by the route legs in the current set.
[0114] The initialize deferral penalties are initialized and only a subset of the route legs' deferral penalties is updated. These are the only two steps in batching that change deferral penalties.
[0115] A batch that exceeds the batch size limit might be returned if adding the route leg set R to the batch B will exceed the predetermined maximum number N of route legs and only if the batch B is non-empty. While rare, it is possible that a route leg can recursively safety-threaten more than the predetermined maximum number N of route legs.
[0116] Referring to FIGS. **1** and **4**, an example of route leg batching is illustrated with respect to autonomous guided container bots **110** travelling along one or more of the container deck **130**DC, picking aisles **130**A and piers **130**BW. It should be understood that batching of route legs for autonomous guided goods bots **262** occurs in a same or similar manner as that of the autonomous guided container bots **110**. In FIG. **4**, there is an active route leg a and its subsequent dummy idling route leg A. There is an active route leg i and its subsequent dummy route leg I. There are seven other route legs B, C, D, E, F, G, H. For exemplary purposes, the predetermined maximum number N of route legs in a batch is set not to exceed three route legs.
[0117] With respect to the safety (e.g., collisions) of a bot of a given route leg, to find the first batch BRL**1** of the sequence BRLS of batches BRL**1**-BRLn, the resolver **120**PR analyzes the route legs whose safety will be threatened by active route legs (i.e., autonomous guided bots **110** of these

threatened route legs may not be able to idle at the source location of the autonomous guided bots **110**). If these threatened route legs are not put into the current batch BRL**1**, the autonomous guided bots **110** of the threatened route legs may not remain safe. For example, an autonomous guided bot of route leg B cannot be safely planned to idle at its current location since the autonomous guided bot of active route leg a is moving towards the current location of route leg B's autonomous guided bot (route leg a is a threat to route leg B). Here, route leg B may be considered as having the largest deferral penalty of the route legs unplanned route legs URL. Thus, the resolver **120**PR assigns route leg B in the first batch BRL**1**. As route leg B may sweep by route leg D's source location and thus threats its safety as well, resolver **120**PR also assigns route D (which has the next largest deferral penalty) in the current batch BRL**1**. As a result, the current batch BRL**1** includes legs B and D.

[0118] With respect to the reachability of destination of a autonomous guided bot **110** of or belonging to a given route leg, the resolver **120**PR also analyzes the route legs for blocked routes. For example, the route leg B passes to and terminates adjacent the entrance of the source aisle of route leg C. If route leg C does not get included in the current batch BRL**1**, an autonomous guided bot **110** of route leg B may stay disposed adjacent the entrance of the source aisle for route leg C (i.e., blocking the aisle entrance and the path of the autonomous guided bot **110** of route leg C out of the aisle). Thus, route leg C (e.g., having a next largest deferral penalty after route leg D) is prioritized to put into the current batch BRL**1**. As such, in this example, route legs B, C, D are assigned to the first batch BRL**1** and the resolver **120**PR employs the priority-based divide and search autonomous transport vehicle path planning process PBDS described herein, to plan the route legs B, C, D (e.g., a time the route legs are executed relative to other route legs).

[0119] With the route legs B, C, D planned (i.e., planned route legs PRL), the time optimal paths and trajectories of these three route legs B, C, D are fixed, and the resolver **120**PR proceeds to collect or otherwise obtain route legs for a second batch BRL**2** of the sequence BRLS of batches BRL**1**-BRLn. As there are five route legs E, F, G, H, I that remain unplanned (i.e., unplanned route legs URL), and given the predetermined maximum number N of route legs in a batch BRL is set to three, the resolver **120**PR expects to pick three of the route legs E, F, G, H, I for inclusion in, so as to form, the second batch BRL**2** of route legs. As there is no route leg in the second batch BRL**2** in the beginning, the resolver **120**PR chooses the most urgent route leg (e.g., after updating the deferral penalties based on planning of route legs B, C, D, the route leg with the largest deferral penalty) of the route legs E, F, G, H, I as the first route leg to be assigned to batch BRL**2**. In the example illustrated in FIG. **6**A, the most urgent route leg of the route legs E, F, G, H, I is route leg G, which is the longest route leg and is more likely to cause starvation under the assumption of all route legs having the same need by times. Here, the resolver **120**PR determines route leg G is in the current batch BRL**2**, and that route leg G may sweep by route leg I's source pose and thus, route leg I is assigned into the current batch BRL**2** as well.

[0120] With respect to delaying of route legs, the resolver **120**PR determines that the arrival time of an autonomous guided bot of route leg H to its destination is affected by the route leg G more than it is affected by route legs E or F. Here, the resolver **120**PR determines that, based on preventing delay, route leg H is assigned to the current batch BRL**2** such that the three route legs of the second batch BRL**2** are route legs G, H, I. The resolver **120**PR employs the priority-based divide and search autonomous transport vehicle path planning process PBDS described herein, to plan the route legs G, H, I (e.g., a time the route legs are executed relative to other route legs).

[0121] With the trajectories and paths of route legs G, H, I planned (e.g., planned route legs PRL) the remaining unplanned route legs URL are route legs E, F. The resolver **120**PR groups route legs E, F into a third batch BRL**3** of route legs and employs the priority-based divide and search autonomous transport vehicle path planning process PBDS described herein, to plan the route legs E, F (e.g., a time the route legs are executed relative to other route legs).

[0122] Referring to FIGS. **1**, **6**B, **7**, and **8**, a threat graph (see FIG. **7**) may be provided or otherwise constructed (FIG. **9**, Block **900**) to facilitate threat determination. Here, a route leg X is considered a threat to route leg Y if a planning of route leg X blocks a planning of route leg Y. The threat relations between a pair of route legs can be one, as noted above, a safety threat, a reachability threat, a delay threat, no threat at all, or a dummy same-bot threat. The hierarchy of the threats from highest threat to lowest threat is the safety threat/dummy-same-bot threat, reachability threat, delay threat, and no threat.

[0123] A safety threat is defined where route let X is planned, route leg Y may not be able to safely stay (e.g., route leg X is a safety threat to route leg Y). The definition of safely staying is that route leg X's reachable region intersects with Y's source region. As an example, where route leg X visits route leg Y's current aisle/pier (see route legs a, B in FIGS. **6**A and **6**B, where route leg a visits route leg B's current aisle/location), route leg X is considered a safety threat to route leg Y. As another example, route leg Y starts from the container transfer deck **130**DC, and route leg X may visit route leg Y's source, route leg X is considered a safety threat to route leg Y.

[0124] With respect to a dummy-same-bot threat, a route leg is a dummy threat to its previous same-bot route leg, and we treat dummy threats as safety threats in analysis. This is because subsequent route legs can only be planned when previous route legs have been planned (e.g., planning subsequent legs first is impossible and this threats validity of planning previous legs).

[0125] A reachability threat is defined where route leg X is scheduled to idle at its source or destination forever, route leg Y may not be able to successfully reach its destination (see route legs B and C in FIGS. **6**A, **6**B where an autonomous guided bot **110** of route leg B blocks egress of an autonomous guided bot **110** of route leg C from exiting the pier). Here, route leg X is a reachability threat on route leg Y.

[0126] A delay threat is defined where two route legs' plans may collide, but one route leg can manage to go to its destination regardless of how the other route leg gets scheduled/planned. In this case, the two route legs are considered delay threats to each other. For example, two route legs come from different railways to two other different railways but they may collide on deck (e.g., see route legs G and H in FIG. **6**A as described above).

[0127] No threat is defined where two route legs belong to different autonomous guided bots **110**, **262** and their plans can never intersect with each other.

[0128] It is noted that in the priority-based divide and search autonomous transport vehicle path planning process PBDS described herein, intermediate destinations of an autonomous guided bot **110**, **262** cannot be the autonomous guided bot's current aisle **130**A/pier **130**BW or route leg destination aisle **130**A/pier **130**BW and thus, autonomous guided bots **110**, **262** will not push another autonomous guided bot **110**, **262** out of an aisle **130**A/pier **130**BW due to demanding that occupied aisle **130**A/pier **130**BW as an intermediate destination. It is also noted that for an active route leg, the priority-based divide and search autonomous transport vehicle path planning process PBDS considers whether the active route leg is a safety threat to other route legs since the other route legs are treated as dynamic obstacles most of time.

[0129] Given the above, threat graphs as illustrated in FIG. **7** may be constructed and employed by the resolver **120**PR where nodes of the threat graphs are route legs and edges are directed and each directed edge from node X to node Y represents that X is a threat to Y. When the threat graph is drawn/generated: solid arrows represent safety threats or dummy-same-bot threats; dashed arrows represent reachability threats; and the nodes in one dashed box have delay threats to each other. The threat graph illustrated in FIG. **7** represents the exemplary routes of FIGS. **6**A and **6**B.

[0130] The threat graph illustrated in FIG. **7** indicates the following: route leg a is a safety threat to route leg B since route leg a sweeps route leg B's source location; route leg A is a safety threat to route leg B since if route leg A is scheduled to move out the aisle, route leg B's source location might be swept by route leg A; route leg B is a reachability threat to route leg C since if route leg B arrives and stays there before route leg C leaves, route leg C cannot reach its destination; route leg F is a reachability threat to route leg E since if route leg F arrives and stays there before route leg E arrives, route leg E cannot reach its destination; route legs A, B, C, F, E are safety threats to route leg D since they all may sweep route leg D's source pose; route legs G and H are safety threats to route leg i since both may sweep by route leg i's source pose; route leg I is a reachability threat to both route legs G and H since route leg I can block both legs if route leg I is planned to idle; route leg G is a safety threat to route leg H since route leg G sweeps by route leg H's source pose; route leg H is a reachability threat to route leg G since if route leg H arrives at its destination earlier, route leg G can only stay; route legs A, B, C, D, E, F are delay threats to each other; route legs G, H, and I are also delay threats to each other; and any pair of route legs across the above delay-threatening group groups has no threat (i.e., route legs I, H, G are not threats to route legs A, B, C, D, E, F.

[0131] Referring to FIGS. **1** and **6**B, calculation of reachable regions of a route leg, by the resolver **120**PR is described, given the route leg source and destination. The reachable regions are employed by the resolver **120**PR to determine threats between route legs.

[0132] An over-approximation reachable region of a route leg should include all the possible reachable regions of all the possible trajectories from the route leg source pose to its destination pose or all possible intermediate destinations. The route leg includes the following bounding regions: the route leg's possible reachable region on the transfer **130**DC given all the possible trajectories; the route leg's possible reachable region on its source aisle **130**A/pier **130**BW given all the possible trajectories (optional for route legs from the transfer deck **130**DC); the route leg's possible reachable region on its destination aisle **130**A/pier **130**BW given all the possible trajectories; and the route leg's possible reachable regions on the aisles **130**A/piers **130**BW of intermediate destinations.

[0133] The resolver **130**PR may exclude the reachable region on intermediate destination railways is excluded because: it is not required for safety or reachability threat check—the intermediate destination cannot be on aisles/piers that are identical as some route leg source aisles/piers or route leg destination aisles/piers and thus it cannot impose a safety threat or reachability threat to any other route leg; and the transfer deck region is enough to indicate a delay threat—if two route legs can visit the same intermediate destinations (between the source and destination), and this makes them delay threats to each other, checking their deck reachable region is enough to find out this relation and checking the intermediate destination aisles/piers may be optional. Here, the resolver **120**PR may be more efficient by excluding the reachable region on intermediate destination railways and only calculating the reachable regions on the transfer deck, source aisle/piers, and destination aisles/piers.

[0134] Referring to route leg E in FIG. **6**B, the source, destination, and intermediate destination of route leg E illustrated. The reachable regions of route leg E on the container transfer deck **130**DC, source pier **130**BW, and destination aisle **130**A are identified as the three bounded regions (i.e., reachable regions ES, ED, EE). Given the reachable regions ES, ED, EE, the resolver **120**PR determines that route leg E may sweep route leg D's source pose and determines route leg E is a safety threat to route leg D.

[0135] Referring also to FIG. **8**, threat relation lemmas will be described.

[0136] Reachability Threat Lemma. If route leg X is at most a reachability threat (i.e., reachability threat, delay threat, no threat) to route leg Y, Y can always stay safe regardless of how X gets scheduled. For example, with reference to the threat graph of FIG. **7**, route leg B is a reachability threat to route leg C, and regardless of how the autonomous guided bot **110** belonging to route leg B moves, route leg C is always safe.

[0137] Delay Threat Lemma. If route leg X is at most a delay threat (i.e., delay threat, no threat) to route leg Y, there always exists a feasible plan for route leg Y to reach its destination regardless of how route leg X gets scheduled. For example, again referring to FIGS. **6**A, **6**B, and **7**, route legs C and E can only collide the container transfer deck **130**DC, which can be avoided by delaying one of route legs C and E.

[0138] No Threat Lemma. If route leg X and route leg Y have not threat, they can be scheduled in parallel.

[0139] Transitivity Lemma. If there exists a path (i.e., a sequence of edges) from node X to node Y and there exists an edge with a threat type THREAT, we say route leg X a at least a THREAT to route leg Y.

[0140] Given a path from route leg X to route leg Y, the threat from route leg X to route leg Y given this path is determined by the least severe threat on this path. Here, if route leg A is a safety threat to route leg B, and route leg B is a delay threat to C, then route leg A is a delay threat to route leg C because route leg A may push route leg B to move but will never invalidate the reachability or safety of route leg C.

[0141] Given all the paths from route leg X to route leg Y, we know the threat from route leg X to route leg Y is the most severe threat among all the paths. Continuing the previous example route leg A is a delay threat to route leg C. If route leg A is a safety threat to route leg D and route leg D is a reachability threat to route leg C, route leg A is a reachability threat to route leg C since route leg A may push route leg D to move around and route leg D may block route leg C from reaching its destination.

[0142] With respect to the deferral penalty, the rules described herein (e.g., the route leg that is most impacted by being deferred into the next batch is included in the current batch) are employed by the resolver **120**PR for selecting route legs to include in a batch BRL of route legs. The impact on a route leg due to deferring the route leg is captured or otherwise accounted for by the deferral penalty.

[0143] Recall that, the objective value of a priority based search is mainly determined by the following metrics: the number of failed legs NF; the number of unresolved conflicts N.sub.UC; and all route leg durations D={d.sub.i}.sub.i. Given all need times T, failed legs weight w.sub.1, unresolvable conflicts weight w.sub.2, the equation of computing the objective value is as follows:

[00001] $F = w_1 N_F + w_2 N_{\mathrm{UC}} + F_D(D,T) + F^*$ [Eq. 1]

[0144] where, F* is a user-specified penalty on de-prioritizing certain route legs (e.g., charge legs, legs from deck), and F.sub.D maps all the route leg duration D and need times T to a penalty F.sub.D(D, T), which includes flow time and the penalty of missing need times:

[00002] $F_D(D,T) = .\mathrm{Math.}_{d_i \in D}\, d_i + .\mathrm{Math.}_{t_j \in T}\, c_t \max(0, -t_j + .\mathrm{Math.}_{d_k \in D \mathrm{and} t_j \mathrm{involves} d_k}\, d_k)^2$ [Eq. 2]

[0145] Considering at each route leg's contribution to the above objective: [0146] 1. Failure and unresolvable collisions: [0147] a. if this route leg ("this route leg" as used in the present disclosures refers to the route leg currently being analyzed for inclusion in a batch of route legs) fails, it contributes w.sub.1, [0148] b. if this route leg has n.sub.i unresolvable collisions, it contributes w.sub.2n.sub.i; [0149] 2. if a route leg takes longer to complete, it also increases the duration penalty (i.e., flow time penalty and the need time penalty that involves this leg); and [0150] 3. user-specified penalty for de-prioritizing this route leg.

[0151] Whether the route leg fails may be dismissed or otherwise skipped as failure can only be true or false, and failure can be reflected in the number of unresolved conflicts. As such, if putting a route leg into the current batch has n.sub.i unresolvable conflicts and duration d.sub.i, and putting a route leg into the next batch has n′.sub.i unresolvable conflicts and duration d′.sub.i, its deferral penalty is defined as follows:

[00003] $= w_2(n_i' - n_i) + (f_D(d_i') - f_D(d_i)) + {}^{*}$ [Eq. 3]

[0152] Where f.sub.D is the duration penalty function with a scope on this leg. The duration penalty function being defined as follows:

[00004] $f_D(d_i) = d_i + .\mathrm{Math.}_{t_j \in T \mathrm{and} t_j \mathrm{involves} d_i}\, c_t \max(0, -t_j + .\mathrm{Math.}_{d_k \in D \mathrm{and} t_j \mathrm{involves} d_k}\, d_k)^2$ [Eq. 4]

[0153] As seen above, the deferral penalty is determined by three metrics: the unresolvable collision number increase (n′.sub.i−n.sub.i), which is related to the safety threat as described herein; two durations d′.sub.i and d.sub.i under two conditions respectively, which are related to the reachability and delay threats as described herein; and user specified penalty Δ* for deferring certain route legs as described herein. Breaking deferral penalty ties will be discussed herein.

[0154] The resolver **120**PR, when employing the priority-based divide and search autonomous transport vehicle path planning process PBDS, is configured to estimate the unresolvable collision number increase (n′.sub.i−n.sub.i) using safety threats. Here, the resolver **120**PR determines the unresolvable collision number increase (n′.sub.i−n.sub.i) by checking how many legs in the current batch BRL safety-threaten a route leg that is to be selected. For example, a route leg B in the current batch BRL of route legs is considered to safety-threaten another route leg A if route leg B has a safety threat edge to route leg A in the threat graph, such as the threat graph illustrated in FIG. **7**.

[0155] The resolver **120**PR, when employing the priority-based divide and search autonomous transport vehicle path planning process PBDS, is also configured to estimate the two durations d′.sub.i and d.sub.i. As described herein, two durations d′.sub.i and d.sub.i estimated for determining the deferral penalty. At lest three methods may be employed by the resolver **120**PR for estimating or otherwise calculating the two durations d′.sub.i and d.sub.i. The three methods discussed below for calculating the two durations d′.sub.i and d.sub.i are discussed in order of accuracy with the least accurate of the methods being discussed first.

[0156] The first method may be referred to as the Naïve method. To evaluate the impact of the reachability threat and the delay threat for a route leg naïvely, the resolver **120**PR is configured to: compute the reference trajectory time of the route leg as d.sub.i; and increase d.sub.i to obtain d′.sub.i where d′.sub.i is equal to .sub.i(d.sub.i+δ1+δ2). Here, δ1 equals 18n.sub.RT:n.sub.RT and is the number of route legs in the current batch of route legs having reachability threats to this route leg, where 18 is a user-specified parameter in seconds, which in other aspects may be larger or smaller than 18 seconds. δ2 equals 3n.sub.DT:n.sub.DT and is the number of route legs in the current batch of route legs having delay threats to this leg (excluding the route legs that already have a reachability threat to this route leg), where 3 is a user-specified parameter in seconds, which in other aspects may be larger or smaller than 3 seconds.

[0157] The second method increases the accuracy of the first method. For example, in the second method the route leg duration d.sub.i is calculated by calling the priority-based divide and search autonomous transport vehicle path planning process PBDS (e.g., the single bot planner) to compute the end time based on all the previously planned route legs; however, d′.sub.i is determined in the same manner as noted above with respect to the first method (i.e., d′.sub.i is equal to .sub.i(d.sub.i+δ1+δ2).

[0158] The third method for determining the two durations d′.sub.i and d.sub.i includes systematically evaluating the effects of reachability threats and delay threats by using pairwise conflict resolution, temporal propagation, and a traffic map as described in U.S. provisional patent application No. 63/631,176 having attorney docket number 1127P017176-US (-#1) filed on Apr. 8, 2024 and titled “System and Method for Priority Based Management of Autonomous Vehicle Fleet,” the disclosure of which is incorporated herein by reference in its entirety.

[0159] The user specified penalty described above with respect to Equation 3 is what may be referred to as a large penalty (e.g., 100,000 seconds; in other aspects the penalty may be larger or smaller than 100,000 seconds) for route legs from the container transfer deck **130**DC or charge route legs to facilitate prioritization of the route legs in an early (formed) batch of route legs. In other aspects, the number of route legs that may be active (e.g., so that an autonomous guided bot belonging to the route leg immediately proceeds to a destination) substantially immediately upon planning of the route legs are prioritized into a batch of route legs by weighting the deferral penalty of these route legs (e.g., with any suitable user-defined weight, such as a weight of 60) in the priority-based divide and search autonomous transport vehicle path planning process PBDS. Here,

the weighting may be added to the deferral penalty of a route leg where: the route leg's trajectory start time given d.sub.i is now; and the autonomous guided bot belonging to the route leg can reach its destination and perform its operation.
[0160] As described above, ties between autonomous guided bot routes when forming batches of route routes BRL may be broken, such as when a new or empty batch is being created. As can be seen in Equation 3, if there is no route leg in the current batch, all the items in the equation are zero except the user specified penalty Δ*. When Δ* is also zero, we may end up with 0 deferral penalty for every route leg (e.g., a tie exists between the route legs that are to be added to the current batch). To break the tie between the route legs one or more of the following tie breaks may be applied to break the tie. [0161] Tie Break 1—where every route leg's deferral penalty is zero, the route legs whose delay is more likely to violate need time constraints (which is given by f.sub.D(d.sub.i)) is picked so as to break the tie; and [0162] Tie Break 2—where Tie Break 1 results in zero penalty, which means all route legs are expected to meet deadlines, a slack time sum of the route legs duration is employed to lead to deadlines. For example, for each route leg duration d.sub.i

[00005] $$\underset{t_j \in T \text{and} t_j \text{involves} d_i}{.\text{Math.}} \left(t_j - \underset{d_k \in D \text{and} t_j \text{involves} d_k}{.\text{Math.}} d_k\right) \quad [\text{Eq}.5]$$

As such, every time the route leg deferral penalty is updated by the resolver, the following values are returned: the deferral penalty (f.sub.D(d.sub.i)); the Tie Break 1 penalty; and in some aspects the Tie Break 2 penalty. To compare (e.g., two or more) the route legs for inclusion into the current batch, the penalty tuples are checked until the tie is broken.
[0163] In the present disclosure, the priority-based divide and search autonomous transport vehicle path planning process PBDS may one or more of: be configured to avoid blocking deferred legs; be configured to run anytime; be configured to adaptively configure batch size; be configured to only evaluate and batch an autonomous guided bot's **110**, **262** first unplanned route leg; be configured to decouple safety threats; and be configured to plan route legs that are executed in parallel (e.g., parallelization).
[0164] To substantially prevent blocking of deferred legs, such as where there is a chance route leg A threats route leg B's reachability, but route leg A is selected into the current batch and route leg B is deferred to future batches. As an example of this threat, route leg A and route leg B go to the same location to perform pick or place, where route leg A is in the current batch and route leg B is deferred. Route leg A may be scheduled to reach its destination and idles there, thus route leg B cannot reach its destination. As another example of this threat, route leg A and route leg B start from the same aisle **130**A or pier **130**BW and route leg B's source location is deeper into the aisle or pier (i.e., further from the transfer deck **130**DC) than route leg A's source location. Here, again, route leg A and route leg B are in different batches. Where route leg A is scheduled to idle at its current location, route leg B cannot leave aisle or pier. This threat may be resolved by placing route legs A and B in the same batch in future planning cycles; however, to substantially avoid blockage of deferred route legs the following method may be employed in the priority-based divide and search autonomous transport vehicle path planning process PBDS by the resolver **120**PR. For a route leg, such as route leg A that threats the reachability of a deferred route leg B the priority-based divide and search autonomous transport vehicle path planning process PBDS collect three boxes (i.e., each box being a spatial boundaries that encompasses the bot's **110**, **262** footprint at a given location on a bot travel surface within the automated storage and retrieval system **100**) from route leg B's specifications. The three boxes are the box (i.e., source box) of the bot **110** belonging to route leg B at the source of route leg B, the box of the bot **110** at the bot's current aisle/pier entrance (this may be the same as the bot's source box if the source of route leg B is at an aisle/pier entrance), and the box (i.e., destination box) of the bot **110** belonging to route leg B at the destination of route leg B. These three boxes are added to route leg A's BoxesPreferredNotIdleAt so that in planning route leg A, the resolver **120**PR does not let the bot of route leg A idle at those

boxes of the bot of route leg B.
[0165] To configure the priority-based divide and search autonomous transport vehicle path planning process PBDS for anytime execution so as to plan route legs in large batches of route legs, the priority-based divide and search autonomous transport vehicle path planning process PBDS is provided with a runtime limit T. Here, the priority-based divide and search autonomous transport vehicle path planning process PBDS when executed by the resolver **120**PR returns currently planned trajectories and paths of route legs in the current batch (even where all route legs of the current batch are not yet planned) when the runtime limit T is met/expires, noting that the trajectories in previously planned batches will not lead to collisions to any unplanned route legs because the previously planned batches are safety-threat-free. Given the runtime limit T, the overall anytime logic, when executed by the resolver **120**PR, is as follows: plan the first batch (noting all the route legs that are safety-threatened by active route legs are collision-free), and keep planning batch by batch where when planning the batches and the runtime exceeds the runtime limit T the trajectories of the route legs in the planned batches are output.
[0166] If a first (or current) batch is too large, one or more of the methods below may be employed to avoid long planning times:
[0167] In a first method, a small (e.g., mini) batch size may be specified, such as by the resolver **120**PR or in any other manner, such as by a user) for the first (or current) batch, or only route legs that are safety threatened by active route legs are planned. The mini first (or current) batch is planned by the resolver **120**PR in a separate thread and the solution of planned route legs is output by the resolver **120**PR if the planning times out (exceeds the runtime limit T) and the full first (or current) batch has not been planned.
[0168] In the second method, with planning each batch (including the first batch), an incumbent solution is recorded by the resolver **120**PR (or other resolver noted herein) in any suitable memory (such as memory **120**M or other memory noted herein) before all the route legs are planned and conflicts are resolved. The route legs in a batch can be further divided into small clusters that exactly cover the full batch. In each cluster, the route legs may have safety threats to each other recursively. For two route legs from different clusters, the route legs in the different clusters do not have safety threats to each other. The trajectories of the route legs are included in a cluster, where: all the route legs in this cluster are planned, the route legs in the cluster have no conflicts to resolve with other route legs in the same cluster, and the route legs in the cluster have no conflicts to resolve with already included clusters' route legs' plans. Here, route leg plans may be extracted from partially planned batches of route legs.
[0169] As noted above, the priority-based divide and search autonomous transport vehicle path planning process PBDS may be configured to adaptively configure the batch size of the route leg batches BRL. As may be realized, choosing a proper batch size for the priority-based divide and search autonomous transport vehicle path planning process PBDS may balance its runtime and performance. While, a large batch size limit can lead to good solution quality planning of route legs in the large batch may take long time. With smaller batch size limits, the priority-based divide and search autonomous transport vehicle path planning process PBDS can plan all route legs very fast but the solution may be of lower quality.
[0170] As may be realized, a selected batch size limit may be employed on different traffic maps, each map having a different number of autonomous guided bots **110**, **262**. To simplify the the priority-based divide and search autonomous transport vehicle path planning process PBDS the batch size limit may be selected so that the batch size limit may be applied to the different traffic maps without creating multiple router builds.
[0171] To adaptively select the batch size limit the resolver **120**PR (or other resolver described herein) may maintain an empirically derived dictionary DIC that maps (i.e., Map Name, Bot Number) to a batch size. For example, non-limiting examples of maps may include: (NBF, 30)=>100; (NBF, 40)=>50; and (NBF, 50)=>30. Here, where there are 35 bots on the NBF map, the

resolver **120**PR selects a batch size limit of 50 route legs.
[0172] Given a current map size, bot number, route leg number, and an estimation of how many conflicts to resolve, the resolver **120**PR (or other resolver described herein) may be configured to predict the total time to plan and adaptively stop collecting route legs into the current batch. Here, the resolver **120**PR is configured in any suitable manner to predict the runtime T given the factors mentioned above.
[0173] The resolver **120**PR (or any other resolver described herein) may be configured to employ a naïve approach for adaptively selecting the batch size limit. Here, the batch size limit is selected on-the-fly. For example, the user inputs into the resolver **120**PR (through any suitable user interface) an initial batch size N (this initial batch size is to be a small number (e.g., about 30 or less than 30), a batch size change step n, and an expected runtime T. With these three parameters N, n, T input to the resolver **120**PR, the resolver executes the priority-based divide and search autonomous transport vehicle path planning process PBDS so as to set the batch size limit to N initially and collect the runtime time of each planning cycle in the past period (e.g., the period being about 5 minutes, although in other aspects the period may be larger or smaller than about 5 minutes). Here, if the average runtime time is smaller than T, the batch size limit is increased by the batch size change step n. If the average runtime time is larger than T, the batch size limit is decreased by the batch size change step n. If planning times out, the batch size limit is reset to the initial batch size N value. This adaptive selective of the batch size limit may also be adaptive to processing power of the resolver, where be applicable to an increased number of the batch size may be decreased in instances of limited processing ability and in other instances, such as where the resolver **120**PR is planning fewer tasks, the batch size may be increased.
[0174] As noted above, the resolver **120**PR, configured with the priority-based divide and search autonomous transport vehicle path planning process PBDS, is configured to evaluate and batch only the first unplanned leg(s) of the bot(s). As described herein, in the priority-based divide and search autonomous transport vehicle path planning process PBDS any route leg with the largest deferral penalty is chosen even if it is not the first unplanned route leg of the autonomous guided bot **110**, **262**. If the selected route leg is not the first unplanned route leg of the autonomous guided bot **110**, **262**, the previous same-bot unplanned legs of the selected route leg are selected via dummy same-bot threats. However, evaluation of all route legs may be avoided as the first unplanned legs are may be prioritized (e.g., relative to subsequent route legs) to pick and plan first. Also, where the second or third unplanned legs of the autonomous guided bot **110**, **262** are selected for planning, the previous legs of the second or third unplanned legs must be selected as well even though they might not be that important to pick. Here, to avoid selection of subsequent route legs, reduce computational times, and increase storage and retrieval system throughput, only the first unplanned route legs of each bot are selected.
[0175] As also noted above, safety threats may be decoupled from the route leg planning process. As described herein, the priority-based divide and search autonomous transport vehicle path planning process PBDS may output batches BRL that exceed the batch size limit (e.g., the batch includes a number of route legs that exceeds the predetermined maximum number of route legs specified for a batch) when too many route legs are safety-threatened by one route leg. Decoupling safety threats from route leg planning may be employed with very small batches (e.g., each batch having a very small number route legs of about 5 route legs or less) where the existence of safety threats may be high and the batch size limit is not to be exceeded (although in other aspects decoupling safety threats may be employed with batches having a batch size limit greater than about route legs). Here, the priority-based divide and search autonomous transport vehicle path planning process PBDS is modified such that with batching of route legs, the resolver **120**PR (or other resolver described herein) does not force selection of route legs that are safety-threatened by the route legs in the current batch. Here, the resolver **120**PR (configured with the priority-based divide and search autonomous transport vehicle path planning process PBDS), when planning a

route leg (such as route leg A), which is in the current batch that threats the safety of a deferred route leg (such as route leg B), reserves route leg B's source box (as described herein) for route leg A and thus route leg A cannot be unsafe to route leg B anymore. It is noted that, safety threats may not be decoupled from active route legs.

[0176] With respect to parallelization, it is noted that the overall strategy of the priority-based divide and search autonomous transport vehicle path planning process PBDS with parallelization (e.g., PBDPS) is to plan all the route legs in a sequence BRLS of batches BRL. Here, the user-defined parameters of the maximum size of a batch to plan (e.g., maximum batch size Nm) and (optionally) the runtime limit T are employed. The resolver **120**PR (or other resolver described herein) determines if the threat graphs (see FIG. **7**) of the route legs to be planned are disconnected. Where the threat graphs are disconnected, the resolver **120**PR launches separate threads to perform the priority-based divide and search autonomous transport vehicle path planning process PBDS for each disconnected part of the threat graphs, and the solutions obtained for each of the separate threats by the priority-based divide and search autonomous transport vehicle path planning process PBDS are returned as a combined solution. The resolver **120**PR initializes non-active route legs' deferral penalties.

[0177] All route legs whose safety is threatened by active route legs are collected/obtained by the resolver **120**PR. Given the collected safety-threatened route legs, the resolver **120**PR initializes a set of batches BRLS where each batch BRL**1**-BRLn contains a set of route legs that at least delay-threaten each other, and route legs across different batches BRL**1**-BRLn do not threaten each other.

[0178] The following are then repeated by the resolver **120**PR, for each of the batches BRL**1**-BRLn in the sequence BRLS (where the batches are analyzed in parallel with each other), until all route legs have been planned or runtime limit has been exhausted, where the solution is returned when the process is terminated. Determine if threat graphs of the batch are disconnected, and where disconnection exists, launch separate threads to perform priority-based divide and search autonomous transport vehicle path planning process PBDS for each disconnected part, where the resolver **120**PR collects the returned solutions from those threads and returns the combined solution. As an option, the resolver **120**PR determines if the runtime has exceeded the runtime limit T, where if the runtime limit T is exceeded the priority-based divide and search autonomous transport vehicle path planning process PBDS process is stopped and a solution is returned (as described herein with respect to the application of the runtime limit T). The resolver **120**PR determines if the number of unplanned route legs is not larger (i.e., is smaller) than the maximum number of route legs in a batch BRL (i.e., the maximum batch size Nm). Where the number of unplanned route legs is smaller than the maximum number of route legs in a batch BRL, all legs are planned and the priority-based divide and search autonomous transport vehicle path planning process PBDS ends. Where the number of unplanned route legs is larger than the maximum batch size Nm, the resolver updates the route leg deferral penalty given the newly added route legs into the batch. The batches BRL are updated where: the unplanned route leg(s) with the highest deferral penalty are added into the corresponding batch; all route legs that are a safety threat to newly added legs are recursively added into the corresponding batch; and batches that are (at least delay) threated by newly added legs are merged. A batch BRL is planned by the priority-based divide and search autonomous transport vehicle path planning process PBDS if this batch's size is not less than the maximum batch size Nm.

[0179] Referring to FIGS. **1**, **6**A, **6**B, and **7** an example route leg planning will be described employing the priority-based divide and search autonomous transport vehicle path planning with parallelization process PBDPS. Here, the maximum batch size Nm is set to 3 route legs in each batch BRL. In the route leg example of FIGS. **6**A and **6**B, whose threat graph is illustrated in FIG. **7**, it is determined by the resolver **120**PR (or other resolver described herein) that the threat graph is disconnected and has two parts **710**, **711**. Here, separate threads (one thread for route legs A, B, C, D, E, F and another thread for route legs I, G, H) are launched by the resolver **120**PR for each part

710, **711**. Planning results are collected from the two threads where the computation of the two threads is effected in parallel by the resolver **120**PR is described below.
[0180] For part **710** of the threat graph it is determined by the resolver **120**PR that there are no disconnected parts to plan in parallel. Non-active route legs' deferral penalty is initialized and sorted to obtain the following order (i.e., route leg, deferral penalty): [0181] B, 20 [0182] C, 18 [0183] E, 12 [0184] F, 8 [0185] D, 5 [0186] A, 0
[0187] The route legs B and D whose safety is threatened by active route leg a are collected. As route legs B and D are threats to each other, only one batch is generated that includes route legs B and D.
[0188] In a first iteration for batch {(B, D)} there are no disconnected parts of the threat graph for batch {(B, D)} to plan in parallel. The number of unplanned route legs is 6, which is greater than the maximum number of route legs **3**. Given newly added route legs B, D, route leg C might be blocked and as such, the deferral penalty of route leg C is increased. Route leg C's current deferral penalty is 18. A placeholder method is employed in this method by simply doubling this penalty, so that the deferral penalty of route leg C is increased to 36. Route leg C is added to the batch with route legs B and D so form batch {(B, D, C)}. There is no need to update the batches and batch {(B, D, C)} is planned.
[0189] In a second iteration for batch {(A, F, E)} the number of unplanned legs is less than the maximum number of route legs in a batch, the all of the route legs are planned, and the thread terminates. The planning of batch {(I, G, H)} may be similar to that of batch {(A, F, E)}.
[0190] Referring to FIGS. **1-8** and **11**, a method includes providing an automated storage and retrieval system **100** (FIG. **11**, Block **1100**) that has: a storage array SA with storage locations **130**S arrayed along aisles **130**A and a static (i.e., non-moving) non-deterministic deck **130**DC communicating with each aisle **130**A; a plurality of autonomous guided bots **110**, each configured for free ranging motion so as to traverse freely along bot paths (e.g., BPT-BPT**3**, see FIGS. **4** and **5**B, as described herein), including optimal paths, on the non-deterministic deck **130**DC so that each autonomous guided bot **110** accesses each storage location **130**S in each aisle **130**A from each location on the non-deterministic deck **130**DC and aisles **130**A; and a controller **120**, **2500** communicably connected to each autonomous guided bot **110** of the plurality of autonomous guided bots **110**. The method also includes assigning, with the controller **120**, **2500**, a series of tasks (FIG. **11**, Block **1110**) to each autonomous guided bot **110**, which series of tasks includes at least one task to at least one autonomous guided bot **110** moving the autonomous guided bot **110** from initial location to a different final location via bot routes **499**A-**499**C (see FIGS. **4**, **6**A and **6**B). The method includes seeking, with a resolver **120**PR, **2500**PR of a bout route planner **120**P, **2500**P of the controller **120**, **2500**, conflicts (FIG. **11**, Block **1120**) between autonomous guided bots **110**, effecting the series of tasks, on the bot routes **499**A-**499**C describing bot paths (see FIGS. **4**, **6**A, and **6**B), and resolving each bot route **499**A-**499**C so as to determine the bot route, at least one bot route **499**A-**499**C being determined based on bot priority. The method includes sequencing, with the resolver **120**PR, **2500**PR, the bot routes **499**A-**499**C (FIG. **11**, Block **1130**) into a sequence of bot route leg batches BRL, wherein route legs (as described herein, and illustrated in the Figures as Manhattan distances, which is a metric in which the distance between two points is the sum of the absolute differences in their Cartesian coordinates), forming each bot route in entirety, are divided into corresponding batch of the sequence of route leg batches BRL.
[0191] In accordance with the present disclosure the method includes one or more of, individually, in any combination with each other, and/or in any combination with the features described above: each bot route **499**A-**499**C is determined batch by batch; the series of tasks include a current task, and at least one of a preceding task and a following task; the at least one task is the current, preceding or following task; the bot route leg batches BRL are prioritized in sequence BRLS, and an earlier or preceding batch in the sequence has a higher priority to a later or subsequent batch in the sequence; at least one bot route leg, in the sequence of bot route legs BRLS describing a bot

route **499**A-**499**C, is deferred to a later bot route leg batch BRL in the sequence of bot route leg batches BRLS; the later bot route leg batch, that includes the deferred at least one bot route leg, is disposed in a later place in the sequence BRLS of bot route leg batches BRL than the at least one bot route leg in sequence of the bot route legs describing the bot route **499**A-**499**C; the deferred at least one bot route leg has a lower priority than undeferred bot route legs; the optimal paths are time optimal paths; the optimal paths are unparameterized paths; and the time optimal paths are time optimal unparameterized paths.

[0192] The following are provided in accordance with the present disclosure and may be employed individually, in any combination with each other, and/or in any combination with the features described above.

[0193] In accordance with the present disclosure an automated storage and retrieval system comprises: a storage array with storage locations arrayed along aisles and a non-deterministic deck communicating with each aisle; a plurality of autonomous guided bots, each configured for free ranging motion so as to traverse freely along bot paths, including optimal paths, on the non-deterministic deck so that each autonomous guided bot accesses each storage location in each aisle from each location on the non-deterministic deck and aisles; and a controller communicably connected to each autonomous guided bot of the plurality of autonomous guided bots so as to assign a series of tasks to each autonomous guided bot, which series of tasks includes at least one task to at least one autonomous guided bot moving the autonomous guided bot from initial location to a different final location via bot routes; wherein the controller is configured with a bot route planner that has a resolver that seeks conflicts between autonomous guided bots, effecting the series of tasks, on the bot routes describing bot paths, and resolves each bot route so as to determine the bot route, at least one bot route being determined based on bot priority, and wherein the resolver is arranged to sequence the bot routes into a sequence of bot route leg batches, wherein route legs, forming each bot route in entirety, are divided into corresponding batch of the sequence of route leg batches.

[0194] In accordance with the present disclosure the automated storage and retrieval system includes one or more of, individually, in any combination with each other, and/or in any combination with the features described above: each bot route is determined batch by batch; the series of tasks include a current task, and at least one of a preceding task and a following task; the at least one task is the current, preceding or following task; the bot route leg batches are prioritized in sequence, and an earlier or preceding batch in the sequence has a higher priority to a later or subsequent batch in the sequence; at least one bot route leg, in the sequence of bot route legs describing a bot route, is deferred to a later bot route leg batch in the sequence of bot route leg batches; the later bot route leg batch, that includes the deferred at least one bot route leg, is disposed in a later place in the sequence of bot route leg batches than the at least one bot route leg in sequence of the bot route legs describing the bot route; the deferred at least one bot route leg has a lower priority than undeferred bot route legs; the optimal paths are time optimal paths; the optimal paths are unparameterized paths; and the time optimal paths are time optimal unparameterized paths.

[0195] In accordance with the present disclosure a method includes providing an automated storage and retrieval system comprising: a storage array with storage locations arrayed along aisles and a non-deterministic deck communicating with each aisle; a plurality of autonomous guided bots, each configured for free ranging motion so as to traverse freely along bot paths, including optimal paths, on the non-deterministic deck so that each autonomous guided bot accesses each storage location in each aisle from each location on the non-deterministic deck and aisles; and a controller communicably connected to each autonomous guided bot of the plurality of autonomous guided bots. The method also includes assigning, with the controller, a series of tasks to each autonomous guided bot, which series of tasks includes at least one task to at least one autonomous guided bot moving the autonomous guided bot from initial location to a different final location via bot routes;

seeking, with a resolver of a bout route planner of the controller, conflicts between autonomous guided bots, effecting the series of tasks, on the bot routes describing bot paths, and resolving each bot route so as to determine the bot route, at least one bot route being determined based on bot priority; and sequencing, with the resolver, the bot routes into a sequence of bot route leg batches, wherein route legs, forming each bot route in entirety, are divided into corresponding batch of the sequence of route leg batches.

[0196] In accordance with the present disclosure the method includes one or more of, individually, in any combination with each other, and/or in any combination with the features described above: each bot route is determined batch by batch; the series of tasks include a current task, and at least one of a preceding task and a following task; the at least one task is the current, preceding or following task; the bot route leg batches are prioritized in sequence, and an earlier or preceding batch in the sequence has a higher priority to a later or subsequent batch in the sequence; at least one bot route leg, in the sequence of bot route legs describing a bot route, is deferred to a later bot route leg batch in the sequence of bot route leg batches; the later bot route leg batch, that includes the deferred at least one bot route leg, is disposed in a later place in the sequence of bot route leg batches than the at least one bot route leg in sequence of the bot route legs describing the bot route; the deferred at least one bot route leg has a lower priority than undeferred bot route legs; the optimal paths are time optimal paths; the optimal paths are unparameterized paths; and the time optimal paths are time optimal unparameterized paths.

[0197] It should be understood that the foregoing description is only illustrative of the aspects of the present disclosure. Various alternatives and modifications can be devised by those skilled in the art without departing from the aspects of the present disclosure. Accordingly, the aspects of the present disclosure are intended to embrace all such alternatives, modifications and variances that fall within the scope of any claims appended hereto. Further, the mere fact that different features are recited in mutually different dependent or independent claims does not indicate that a combination of these features cannot be advantageously used, such a combination remaining within the scope of the aspects of the present disclosure.

Claims

1. An automated storage and retrieval system comprising: a storage array with storage locations arrayed along aisles and a non-deterministic deck communicating with each aisle; a plurality of autonomous guided bots, each configured for free ranging motion so as to traverse freely along bot paths, including optimal paths, on the non-deterministic deck so that each autonomous guided bot accesses each storage location in each aisle from each location on the non-deterministic deck and aisles; and a controller communicably connected to each autonomous guided bot of the plurality of autonomous guided bots so as to assign a series of tasks to each autonomous guided bot, which series of tasks includes at least one task to at least one autonomous guided bot moving the autonomous guided bot from initial location to a different final location via bot routes; wherein the controller is configured with a bot route planner that has a resolver that seeks conflicts between autonomous guided bots, effecting the series of tasks, on the bot routes describing bot paths, and resolves each bot route so as to determine the bot route, at least one bot route being determined based on bot priority, and wherein the resolver is arranged to sequence the bot routes into a sequence of bot route leg batches, wherein route legs, forming each bot route in entirety, are divided into corresponding batch of the sequence of route leg batches.

2. The automated storage and retrieval system of claim 1, wherein each bot route is determined batch by batch.

3. The automated storage and retrieval system of claim 1, wherein the series of tasks include a current task, and at least one of a preceding task and a following task.

4. The automated storage and retrieval system of claim 3, wherein the at least one task is the

current, preceding or following task.
5. The automated storage and retrieval system of claim 1, wherein the bot route leg batches are prioritized in sequence, and an earlier or preceding batch in the sequence has a higher priority to a later or subsequent batch in the sequence.
6. The automated storage and retrieval system of claim 1, wherein at least one bot route leg, in the sequence of bot route legs describing a bot route, is deferred to a later bot route leg batch in the sequence of bot route leg batches.
7. The automated storage and retrieval system of claim 6, wherein the later bot route leg batch, that includes the deferred at least one bot route leg, is disposed in a later place in the sequence of bot route leg batches than the at least one bot route leg in sequence of the bot route legs describing the bot route.
8. The automated storage and retrieval system of claim 6, wherein the deferred at least one bot route leg has a lower priority than undeferred bot route legs.
9. The automated storage and retrieval system of claim 1, wherein the optimal paths are time optimal paths.
10. The automated storage and retrieval system of claim 1, wherein the optimal paths are time unparameterized paths.
11. A method comprising: providing an automated storage and retrieval system comprising: a storage array with storage locations arrayed along aisles and a non-deterministic deck communicating with each aisle; a plurality of autonomous guided bots, each configured for free ranging motion so as to traverse freely along bot paths, including optimal paths, on the non-deterministic deck so that each autonomous guided bot accesses each storage location in each aisle from each location on the non-deterministic deck and aisles; and a controller communicably connected to each autonomous guided bot of the plurality of autonomous guided bots; assigning, with the controller, a series of tasks to each autonomous guided bot, which series of tasks includes at least one task to at least one autonomous guided bot moving the autonomous guided bot from initial location to a different final location via bot routes; seeking, with a resolver of a bout route planner of the controller, conflicts between autonomous guided bots, effecting the series of tasks, on the bot routes describing bot paths, and resolving each bot route so as to determine the bot route, at least one bot route being determined based on bot priority; and sequencing, with the resolver, the bot routes into a sequence of bot route leg batches, wherein route legs, forming each bot route in entirety, are divided into corresponding batch of the sequence of route leg batches.
12. The method of claim 11, wherein each bot route is determined batch by batch.
13. The method of claim 11, wherein the series of tasks include a current task, and at least one of a preceding task and a following task.
14. The method of claim 13, wherein the at least one task is the current, preceding or following task.
15. The method of claim 11, wherein the bot route leg batches are prioritized in sequence, and an earlier or preceding batch in the sequence has a higher priority to a later or subsequent batch in the sequence.
16. The method of claim 11, wherein at least one bot route leg, in the sequence of bot route legs describing a bot route, is deferred to a later bot route leg batch in the sequence of bot route leg batches.
17. The method of claim 16, wherein the later bot route leg batch, that includes the deferred at least one bot route leg, is disposed in a later place in the sequence of bot route leg batches than the at least one bot route leg in sequence of the bot route legs describing the bot route.
18. The method of claim 16, wherein the deferred at least one bot route leg has a lower priority than undeferred bot route legs.
19. The method of claim 11, wherein the optimal paths are time optimal paths.
20. The method of claim 11, wherein the optimal paths are time unparameterized paths.